VoxLat — figure and table gallery

Graded TPMS lattice cooling jacket: homogenization, finite-gap correction, device model, multi-objective optimization, PicoGK export

40 figures, 7 tables · built 2026-10-16 18:00 UTC

`Manufacturable slots optimum: results/designs/task12_slots_best_at_cap_gyroid.json`

Reference problem

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Geometry

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F03 asf convergence — `task1_asf_convergence.png`
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Homogenization

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F31 baseline fronts — `task10_fronts.png`
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RQ3 optimizers

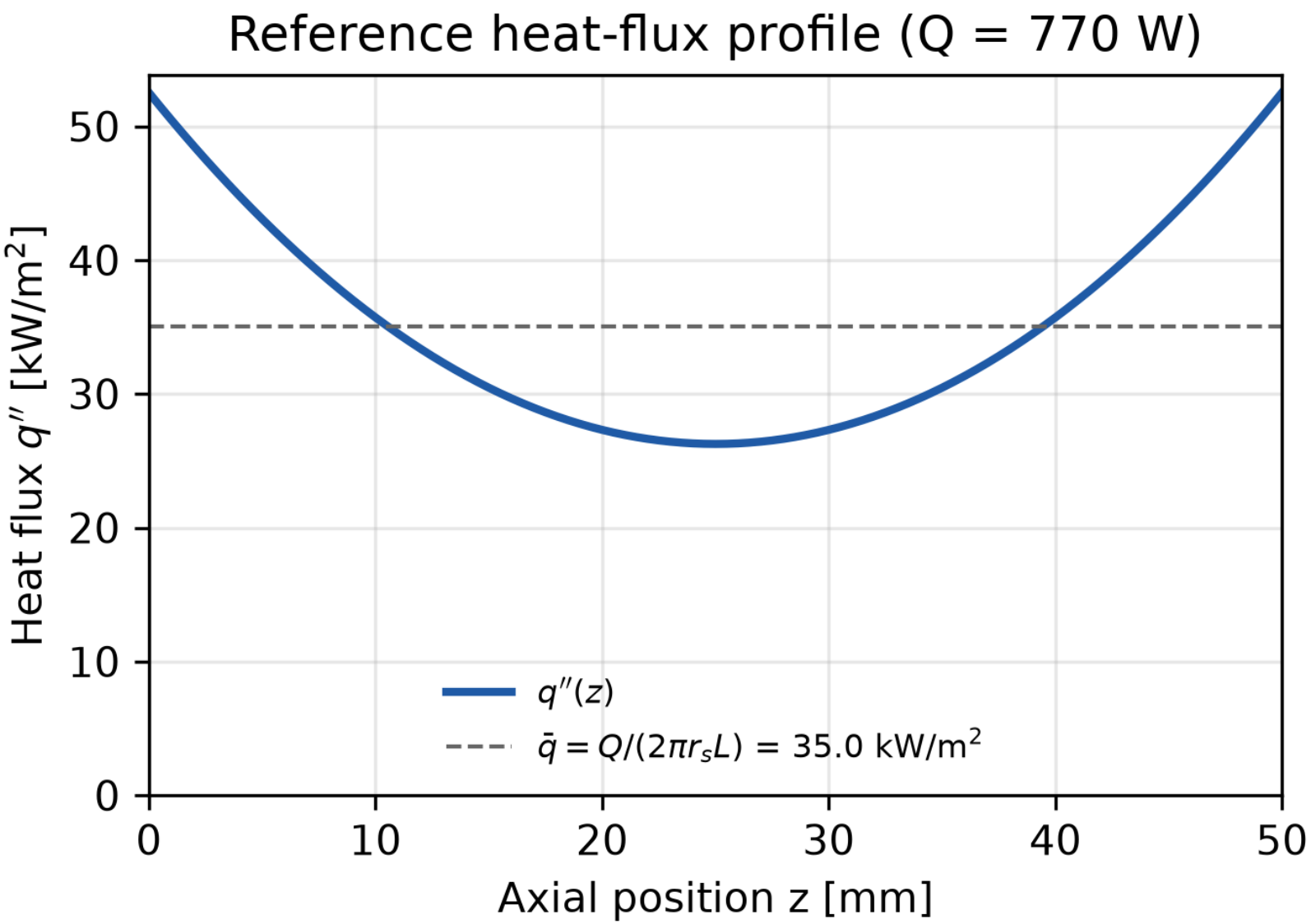
F33 hypervolume — `task11_hypervolume.png`

RQ2 designs

F34 pareto fronts — `task11_pareto_fronts.png`
F35 designs slots — `task11_designs_slots.png`
F36 designs end rings — `task11_designs_end_rings.png`

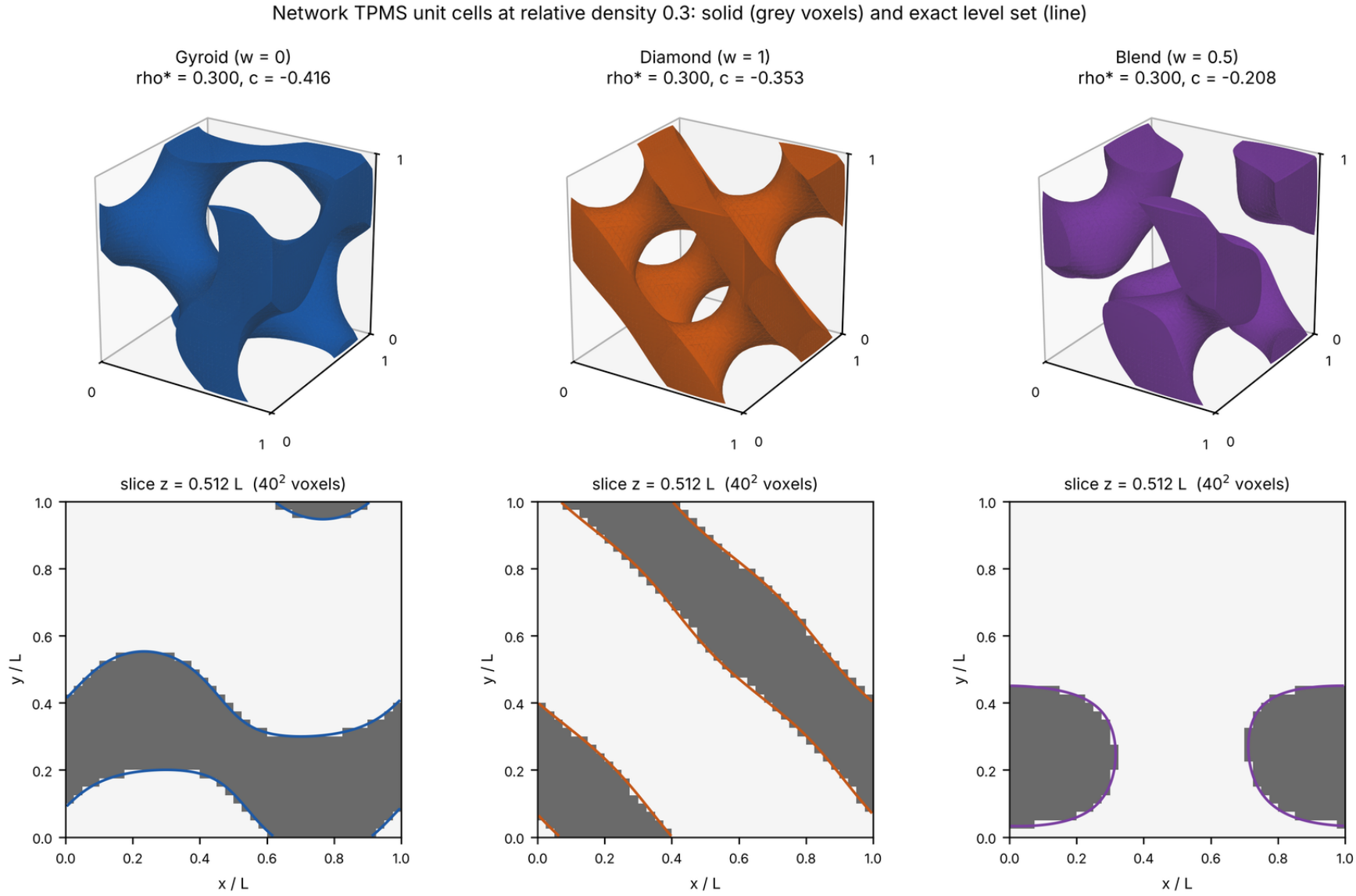
Geometry export	
F37 export renders	`task12_renders.png`
F38 export phase field	`task12_phase_field.png`
F39 export sector slices	`task12_sector_slices.png`
F40 export volume fraction	`task12_volume_fraction.png`
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Closures	
T02 surrogate accuracy	`T02_surrogate_accuracy.csv`
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T04 literature closures	`T04_literature_closures.csv`
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RQ2 designs	
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Geometry export	
T09 export checks	`T09_export_checks.csv`

Reproduce: python scripts/experiments/run_all.py --replot (see README, 'Reproduce the paper')



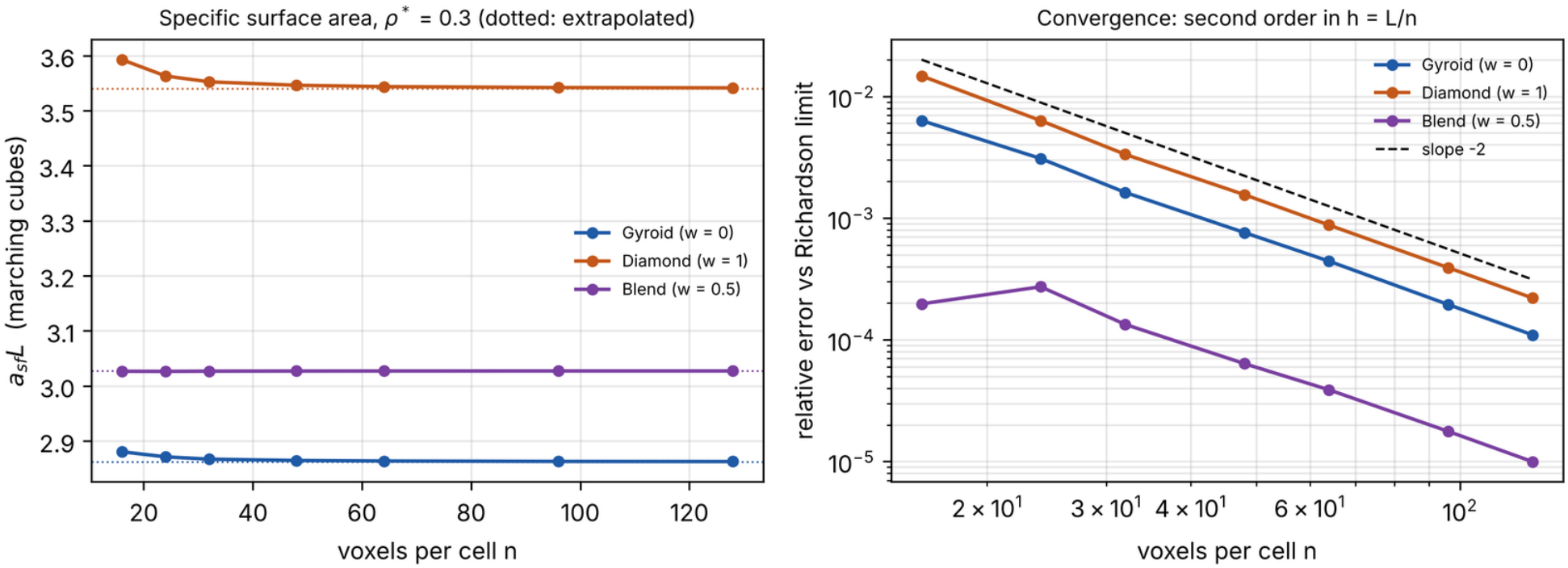
F01. Axial stator heat-flux profile q''(z) of the reference 15 kW motor: mean 35 kW/m^2 (Q = 770 W over the 70 mm x 50 mm sleeve), parabolic with the ends at twice the centre value; integrates to Q within 0.1 %.

File: results/figures/task0_heat_flux_profile.png
Made by: scripts/task0_heat_flux_profile.py (full run: 5 s)



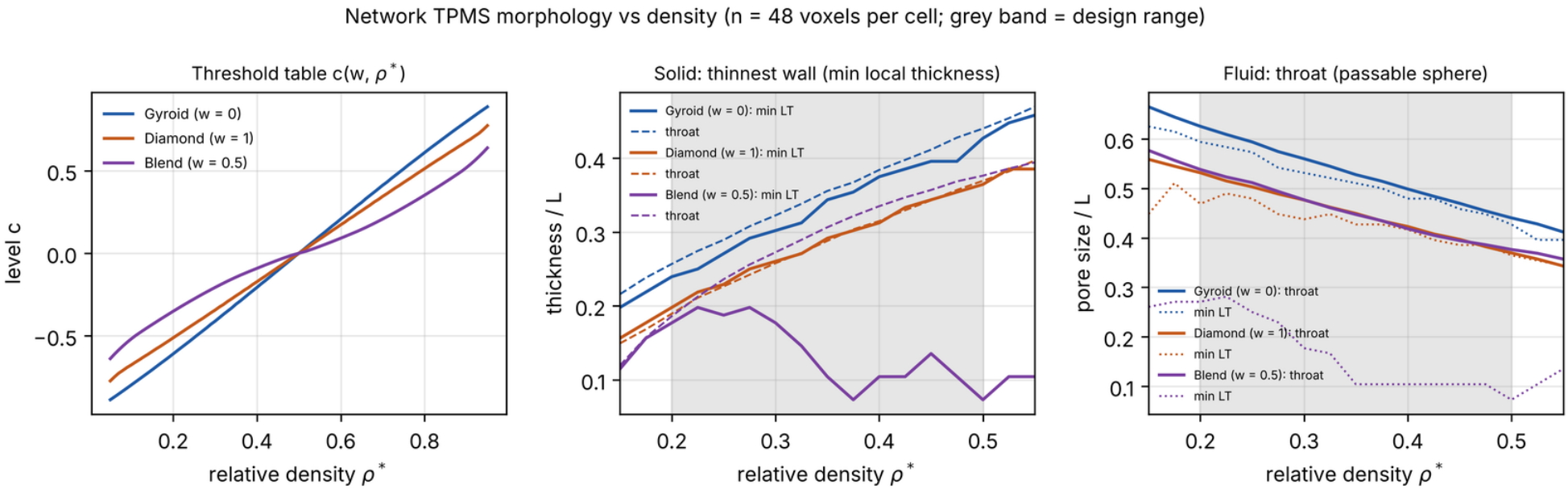
F02. Network TPMS unit cells at relative density 0.30: gyroid (w = 0), 50/50 blend (w = 0.5) and Schwarz diamond (w = 1), rendered and sliced; solid where the blended level set phi_w <= c(w, rho*).

File: results/figures/task1_tpms_cells.png
Made by: scripts/task1_tpms_figures.py (full run: 1-2 min)



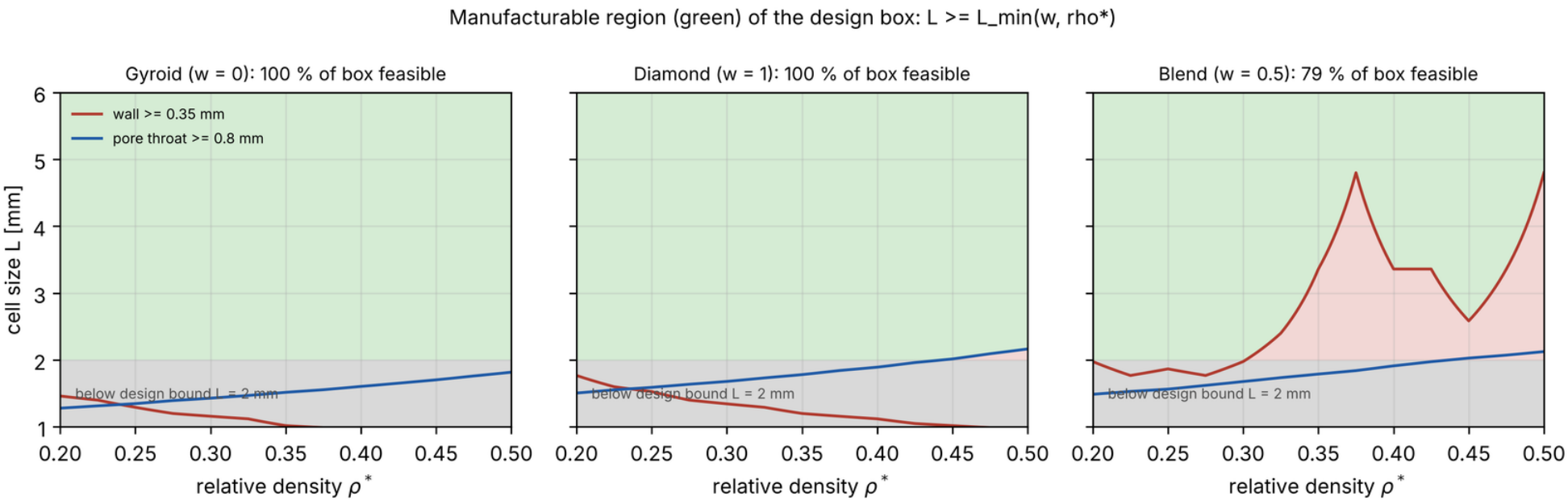
F03. Grid convergence of the specific surface area a_sf L (marching cubes on the periodic level set) for n = 24-64 voxels per cell; the gyroid sheet limit approaches the minimal-surface value ~3.09.

File: results/figures/task1_asf_convergence.png
Made by: scripts/task1_tpms_figures.py (full run: 1-2 min)
Data: results/task1_asf_convergence.csv



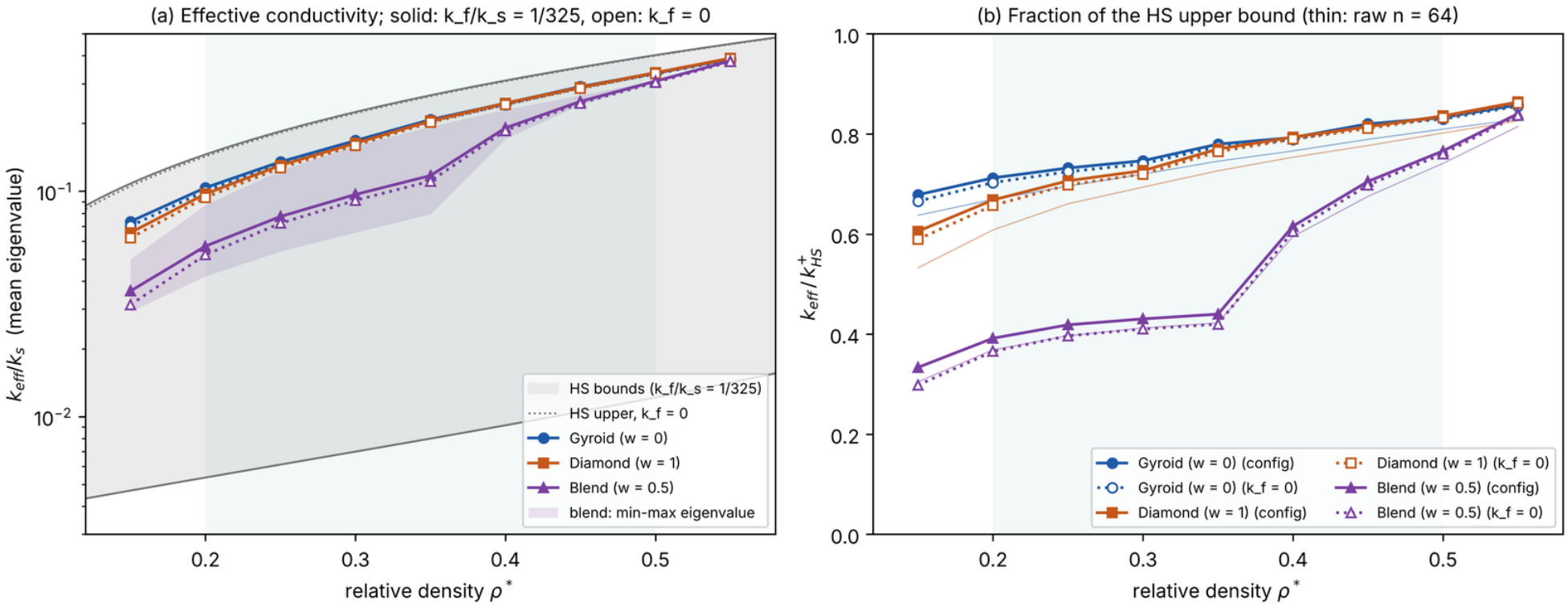
F04. Minimum wall thickness, minimum pore size and surface area vs relative density for gyroid, diamond and blends (n = 48 periodic distance transform). Blends develop pinch-point necks (non-monotonic minimum wall).

File: results/figures/task1_morphology_vs_density.png
Made by: scripts/task1_tpms_figures.py (full run: 1-2 min)



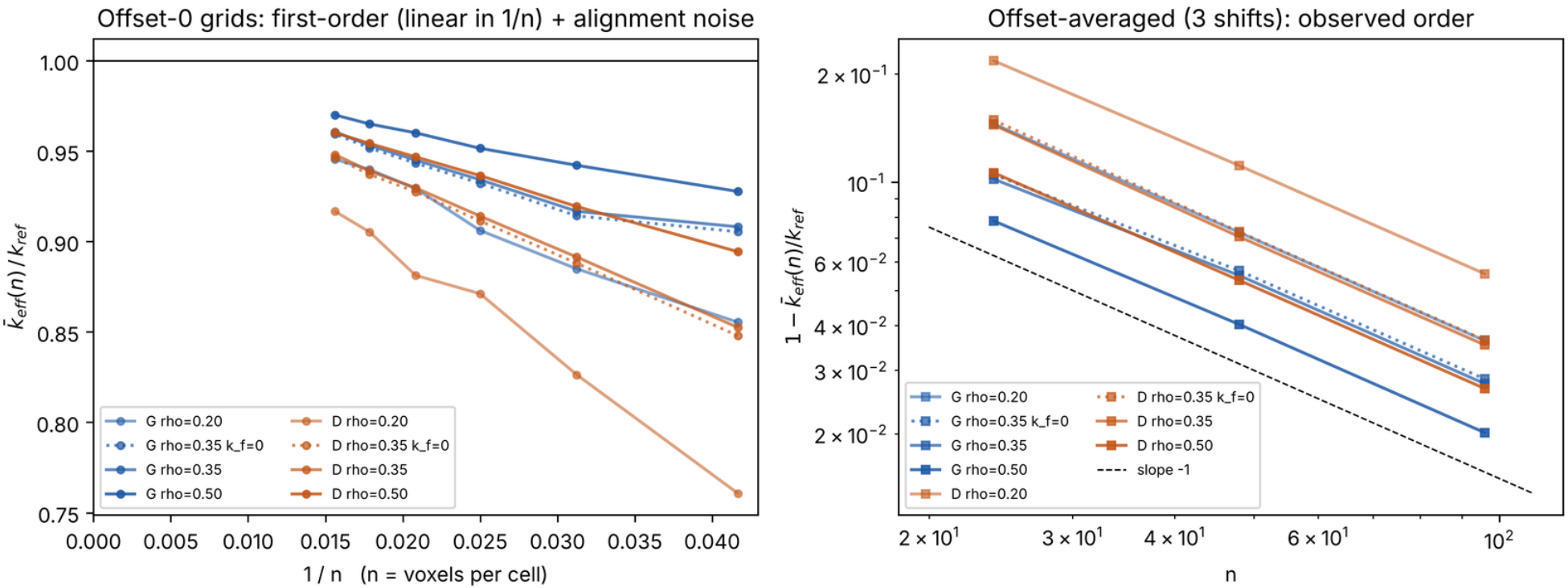
F05. Manufacturable (rho*, L) region per blend weight w for min wall 0.35 mm and min pore 0.8 mm: the cell-size floor L_min(w, rho*) used as the optimization constraint.

File: results/figures/task1_feasible_region.png
Made by: scripts/task1_tpms_figures.py (full run: 1-2 min)



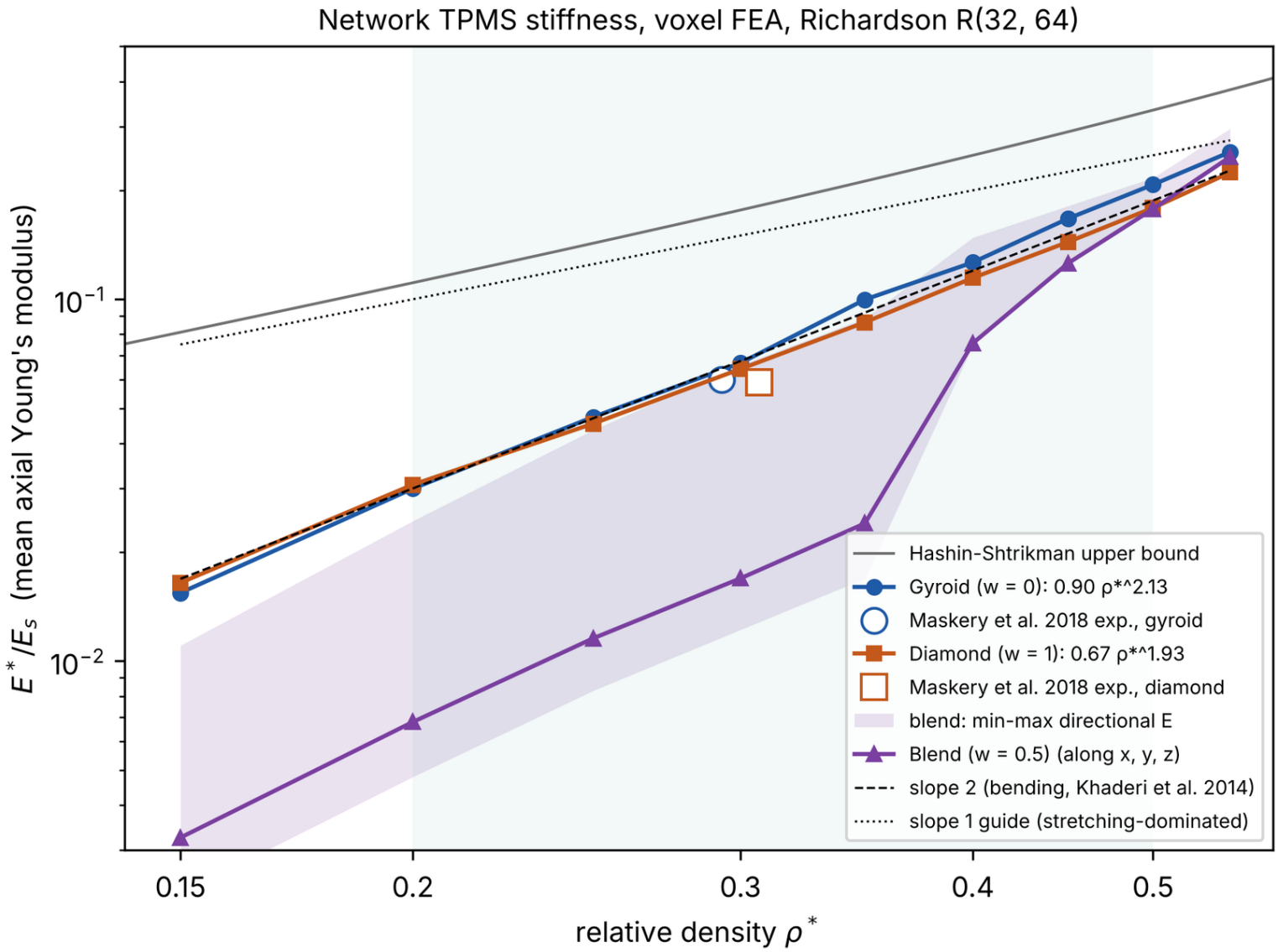
F06. Effective conductivity k_eff/k_s vs relative density for gyroid, diamond and a 50/50 blend (k_f/k_s = 1/325 and k_f = 0), with Hashin-Shtrikman bounds; panel (b) fraction of the HS upper bound. Two-grid Richardson estimator R(32, 64).

File: results/figures/task2_keff_vs_density.png
Made by: scripts/task2_keff_vs_density.py (full run: ~5 min (2 cores))
Data: results/task2_keff_vs_density.csv



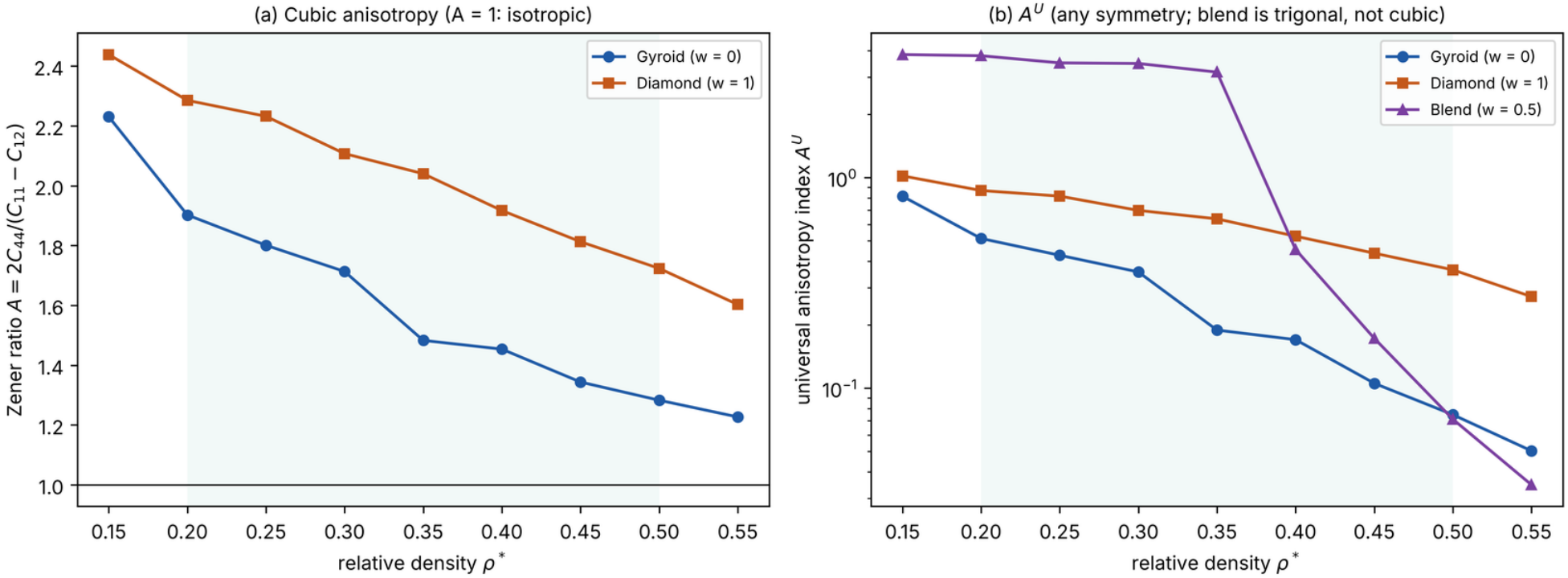
F07. Grid convergence of k_eff: offset-0 grids vs 1/n and offset-averaged error (log-log, slope -1); basis for the R(32, 64) production estimator.

File: results/figures/task2_convergence.png
Made by: scripts/task2_convergence.py (full run: ~15 min (2 cores))
Data: results/task2_convergence.csv, results/task2_convergence_summary.csv



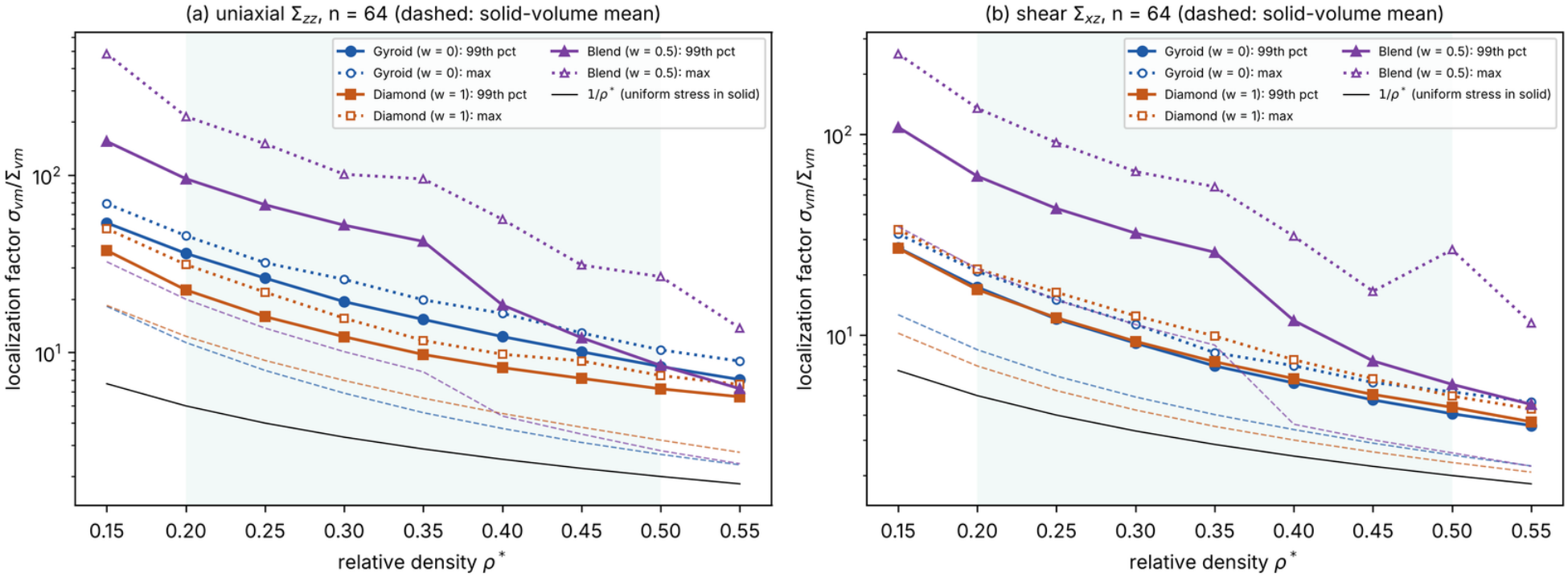
F08. Effective Young's modulus E*/E_s vs relative density (log-log) from periodic voxel FEA, with power-law fits E*/E_s = C rho*^m, the HS bound, a slope-2 guide and published gyroid/diamond data (Maskery et al.); blend shown as its directional range.

File: results/figures/task3_youngs_vs_density.png
Made by: scripts/task3_stiffness_vs_density.py (full run: ~16 min (2 cores))
Data: results/task3_stiffness_vs_density.csv, results/task3_powerlaw_fits.csv



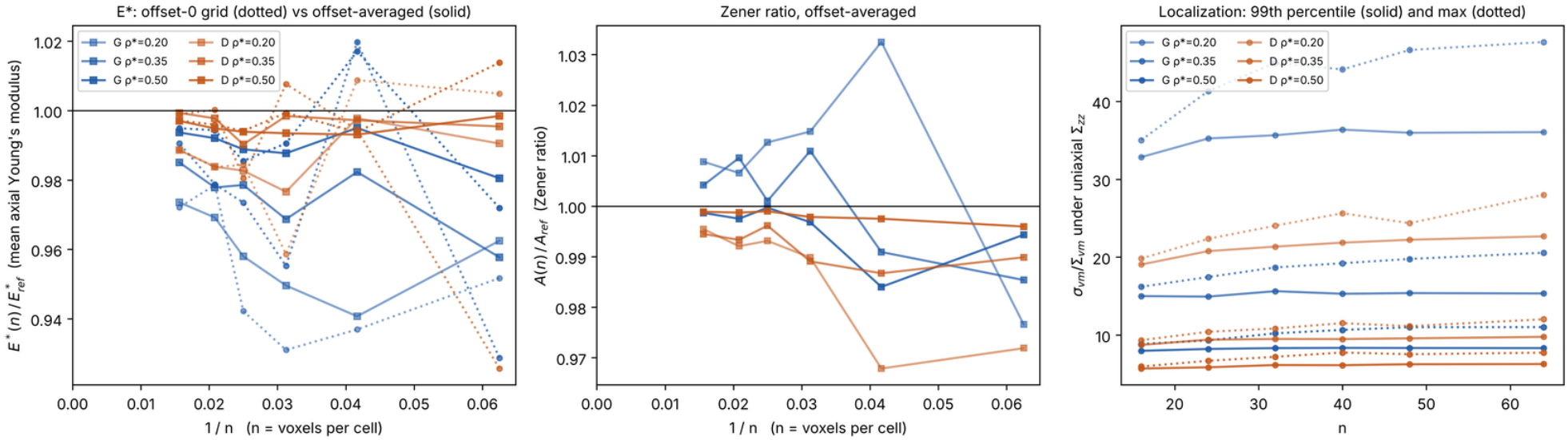
F09. Elastic anisotropy vs relative density: Zener ratio of the cubic gyroid and diamond and the universal anisotropy index A^U (including the trigonal blend).

File: results/figures/task3_anisotropy_vs_density.png
Made by: scripts/task3_stiffness_vs_density.py (full run: ~16 min (2 cores))



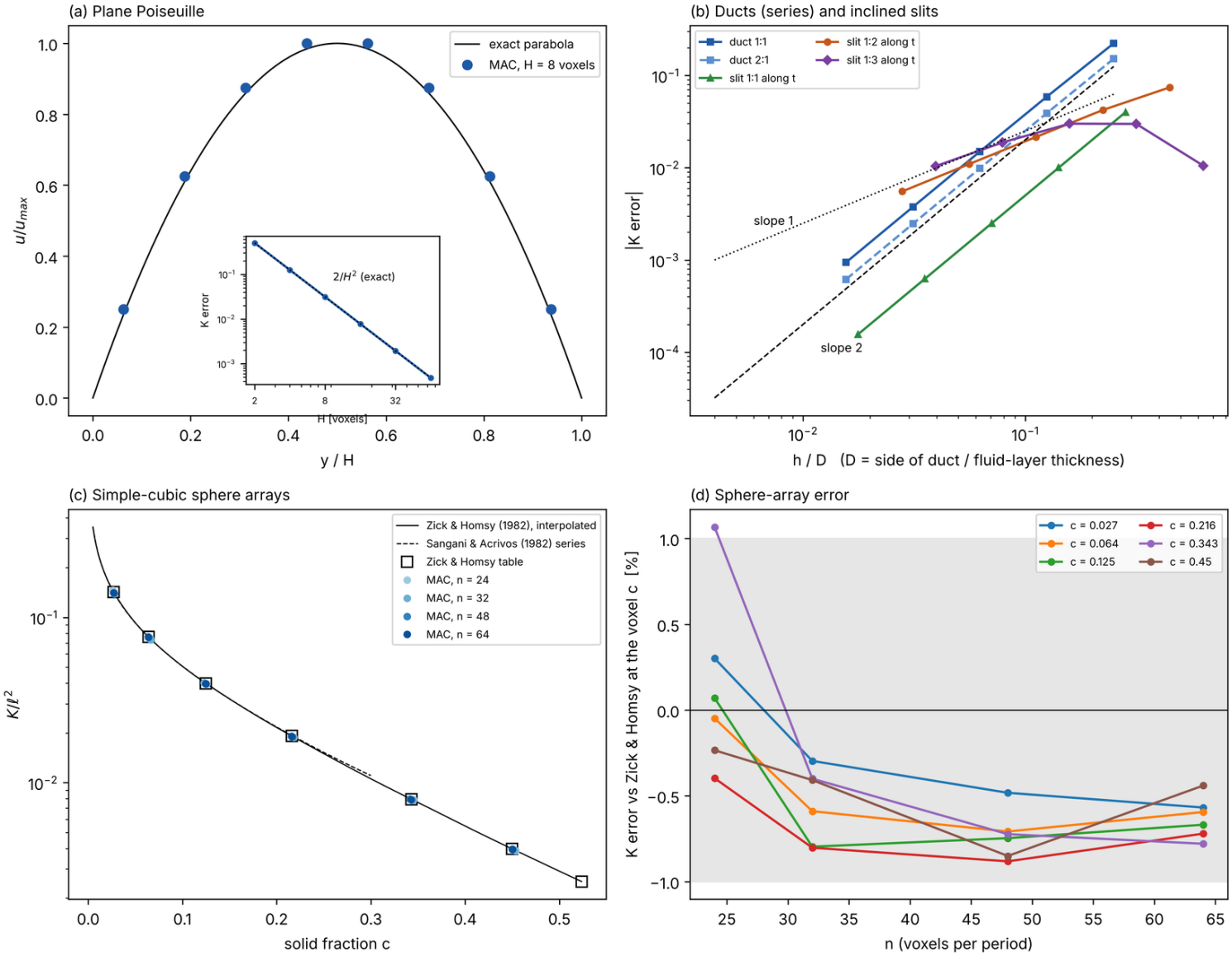
F10. Stress localization factors (99th percentile, maximum and mean von Mises over the macroscopic stress) under unit uniaxial-z and shear-xz stress vs relative density (n = 64).

File: results/figures/task3_localization_vs_density.png
Made by: scripts/task3_stiffness_vs_density.py (full run: ~16 min (2 cores))



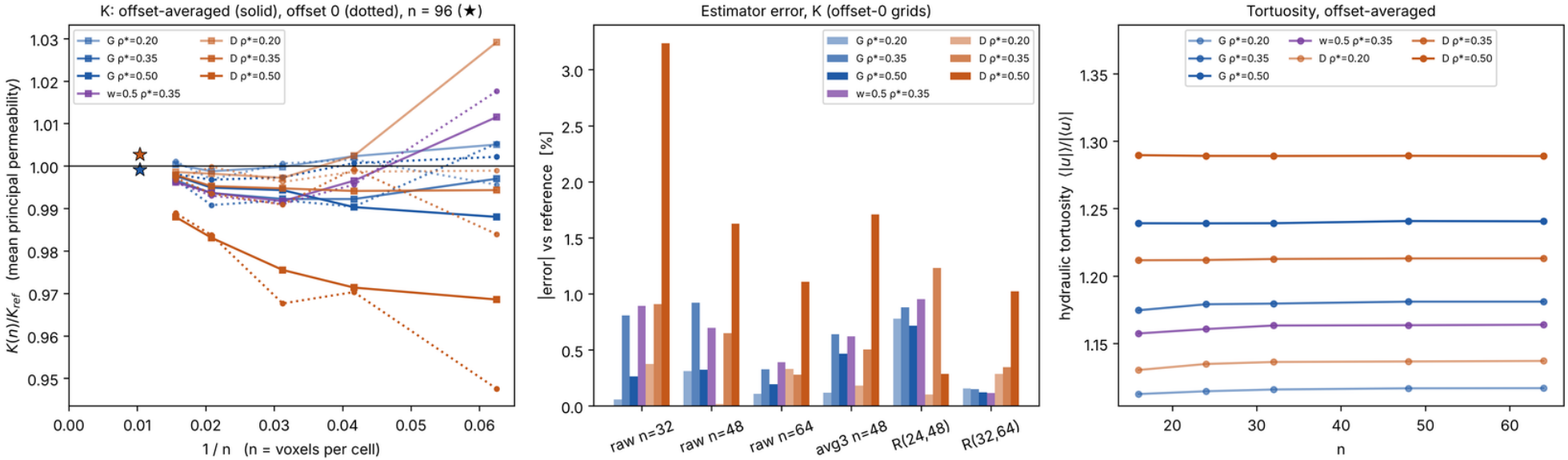
F11. Grid convergence of C_eff and stress localization (n = 16-64, 4 grid offsets each; n = 96 anchors): systematic staircase error vs alignment noise.

File: results/figures/task3_convergence.png
Made by: scripts/task3_convergence.py (full run: ~14 min (2 cores))
Data: results/task3_convergence.csv, results/task3_convergence_summary.csv, results/task3_n96_anchor.csv



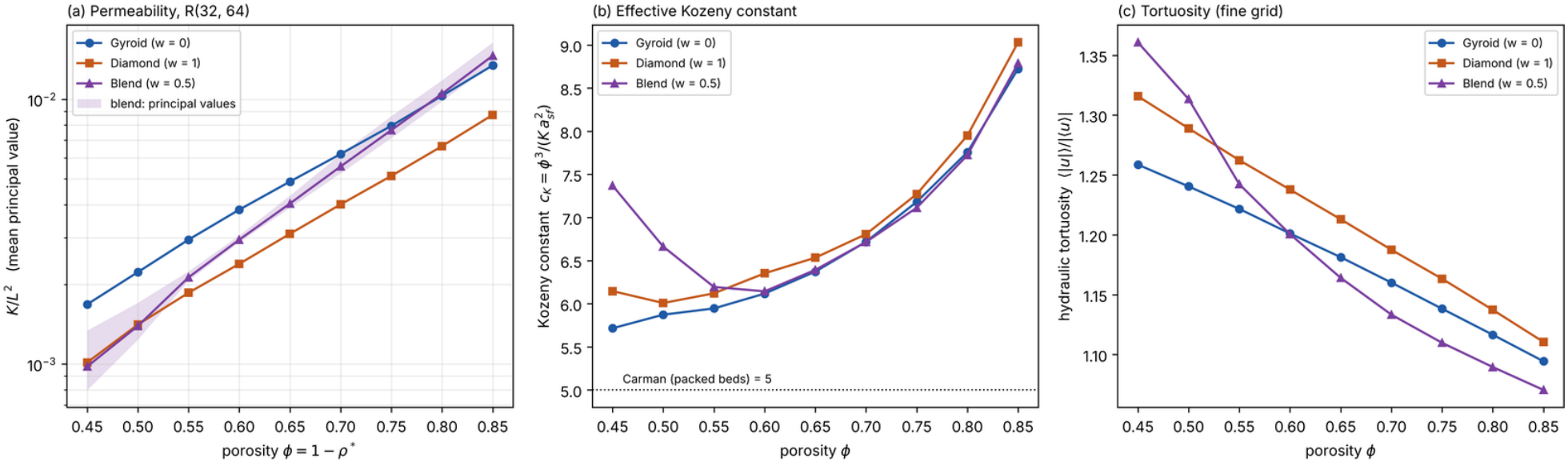
F12. Verification of the MAC Stokes solver: (a) plane Poiseuille profile and 2/H^2 error, (b) square / rectangular ducts and inclined slits vs the series solution, (c) simple-cubic sphere arrays vs Zick & Homsy (1982) and Sangani & Acrivos, (d) sphere error vs resolution.

File: results/figures/task4_verification.png
Made by: scripts/task4_verification.py (full run: ~2 min (2 cores))
Data: results/task4_verification_channels.csv, results/task4_verification_spheres.csv



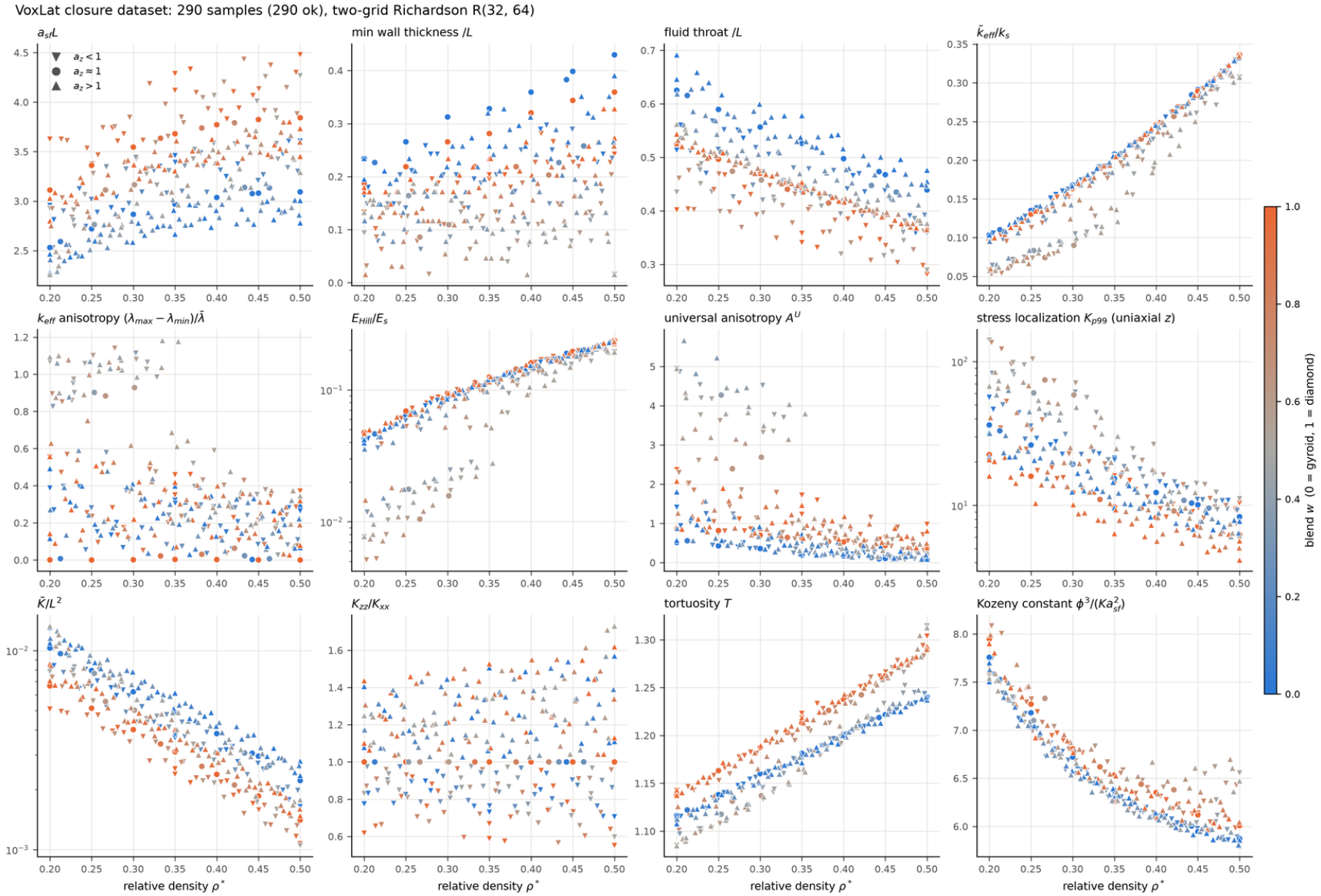
F13. Grid convergence of the permeability K(n)/K_ref vs 1/n with n = 96 anchors, estimator errors and tortuosity.

File: results/figures/task4_convergence.png
Made by: scripts/task4_convergence.py (full run: ~11 min (2 cores, 1.5 GB))
Data: results/task4_convergence.csv, results/task4_convergence_summary.csv



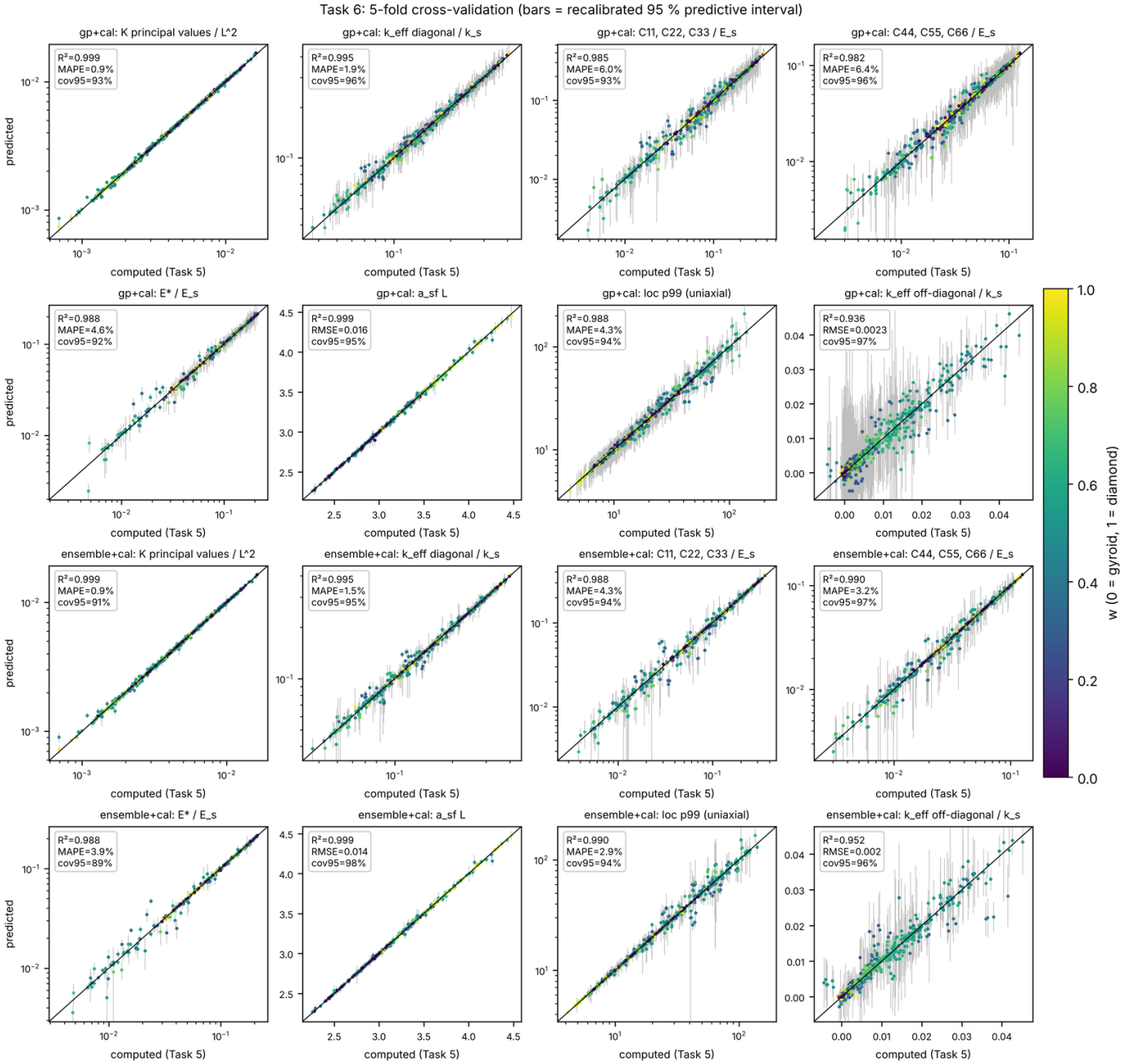
F14. Permeability K/L^2 vs porosity for gyroid, diamond and the 50/50 blend (principal-value band), Kozeny constant and tortuosity; R(32, 64) estimator.

File: results/figures/task4_permeability_vs_porosity.png
Made by: scripts/task4_permeability_vs_porosity.py (full run: ~9 min (2 cores))
Data: results/task4_permeability_vs_porosity.csv



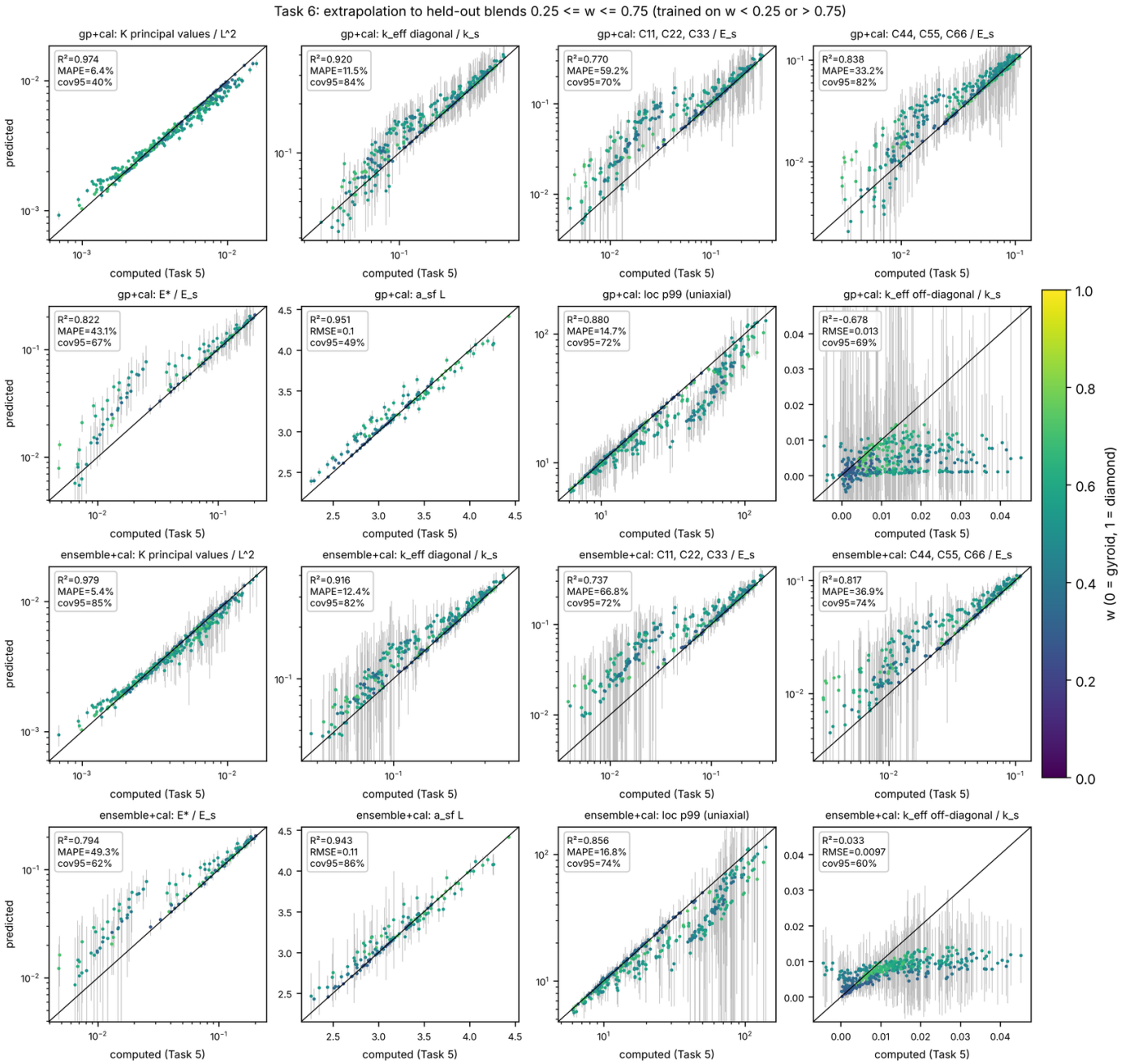
F15. Closure dataset (Task 5; 290 design points theta = (w, rho*, a_z), Sobol + boundary points): every closure vs relative density, coloured by blend weight w.

File: results/figures/task5_closures_quicklook.png
Made by: scripts/build_dataset.py (full run: ~7.4 h (laptop, 4 workers; 27.8 CPU-h))
Data: data/closures.parquet



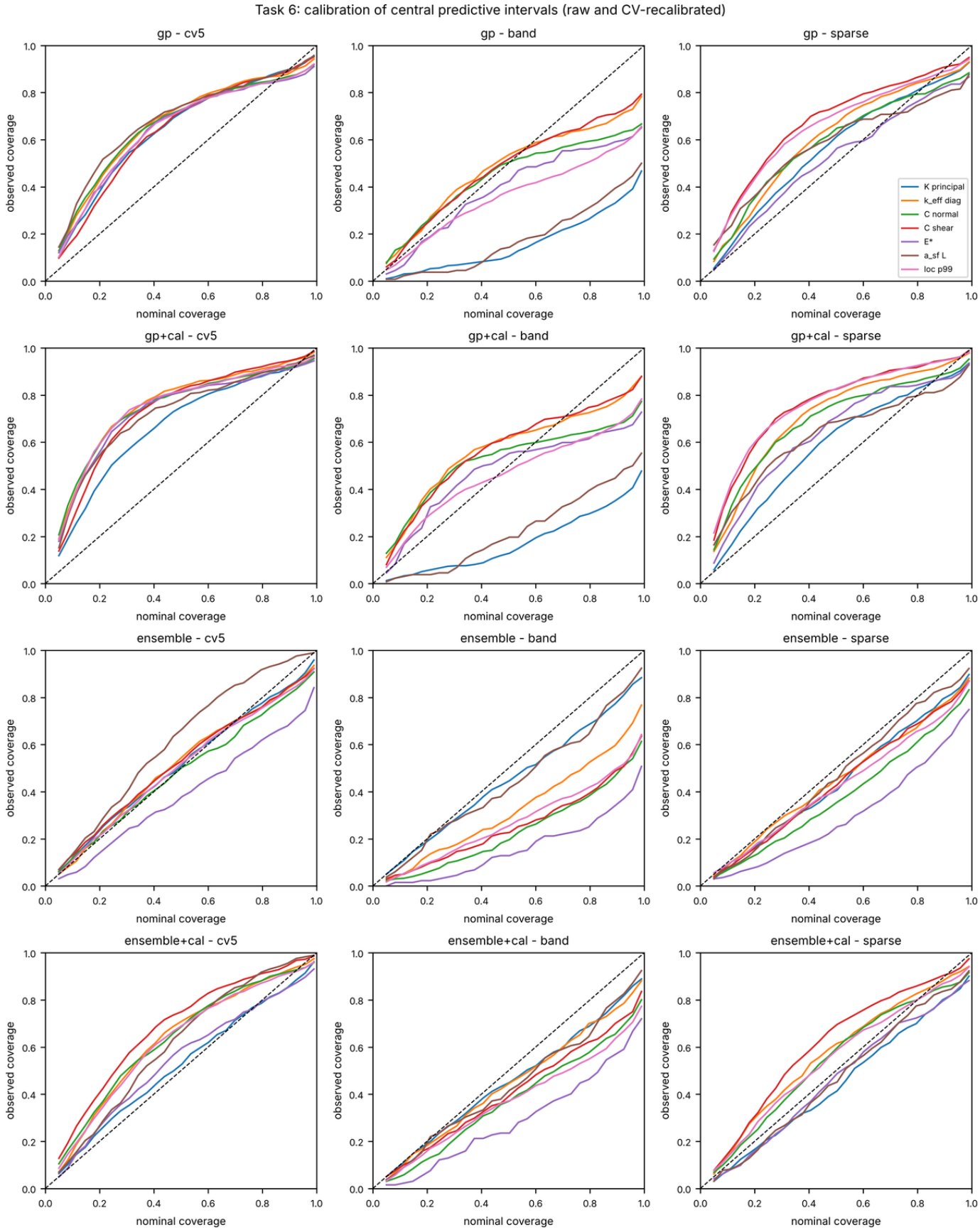
F16. Five-fold cross-validated parity of the GP and deep-ensemble closure surrogates with recalibrated 95 % intervals.

File: results/figures/task6_parity_cv.png
Made by: scripts/task6_evaluate.py (full run: ~10 min (2 cores))
Data: results/task6_metrics.csv, results/task6_predictions.csv.gz



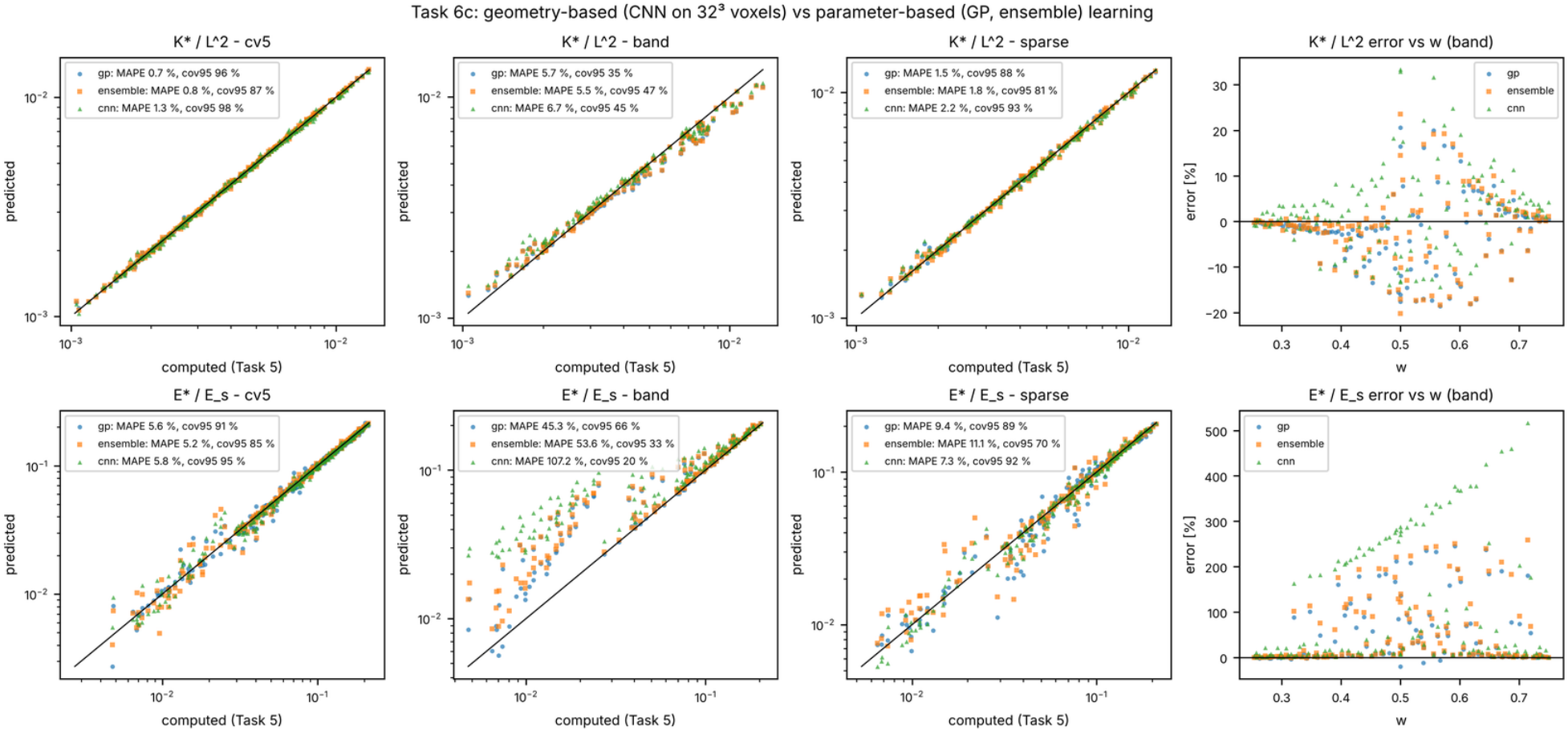
F17. Parity on the blend-band extrapolation split (training on near-pure morphologies, testing on blends).

File: results/figures/task6_parity_band.png
Made by: scripts/task6_evaluate.py (full run: ~10 min (2 cores))



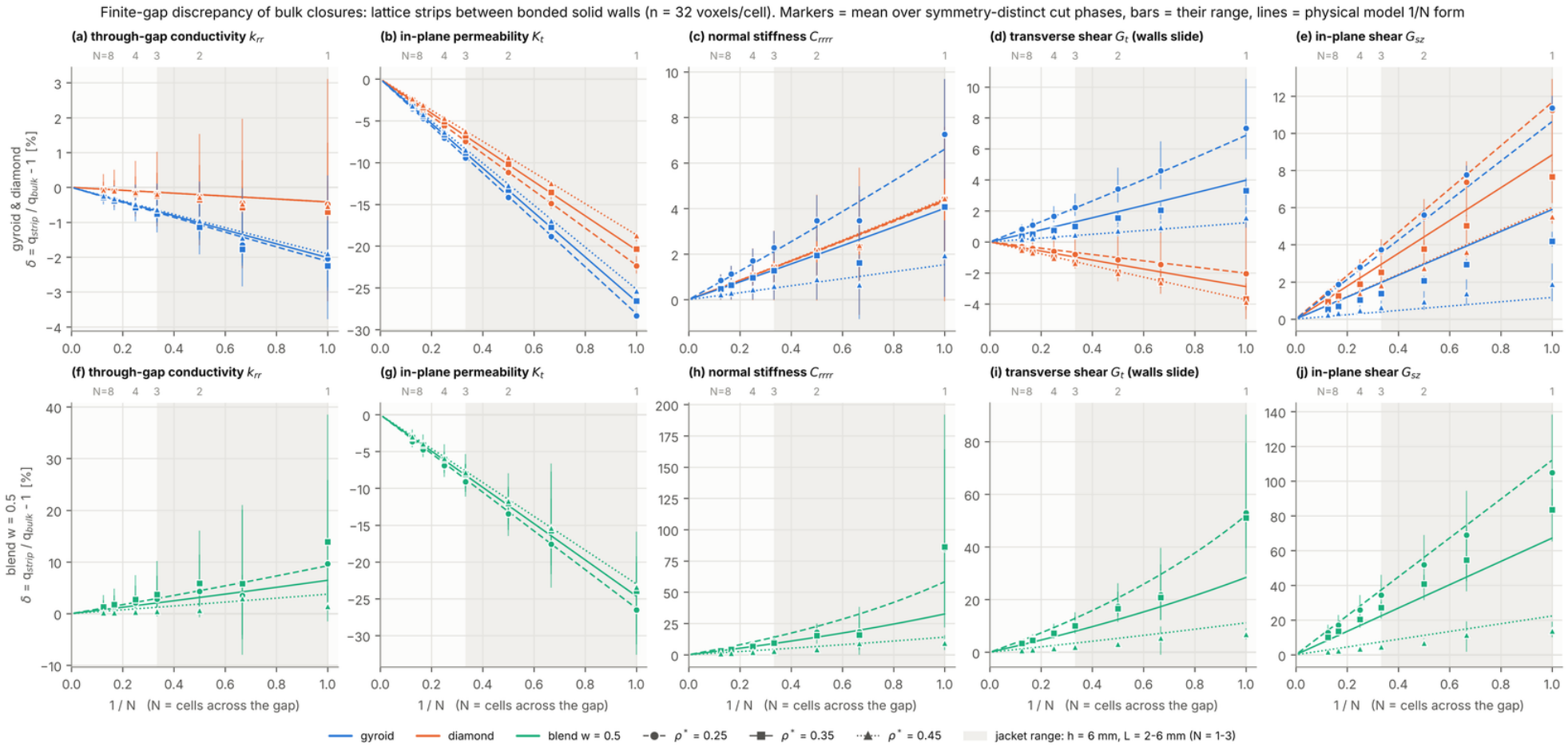
F18. Calibration of the surrogate uncertainty: observed vs nominal interval coverage, raw and recalibrated, for the cv5, band and sparse splits.

File: results/figures/task6_calibration.png
Made by: scripts/task6_evaluate.py (full run: ~10 min (2 cores))



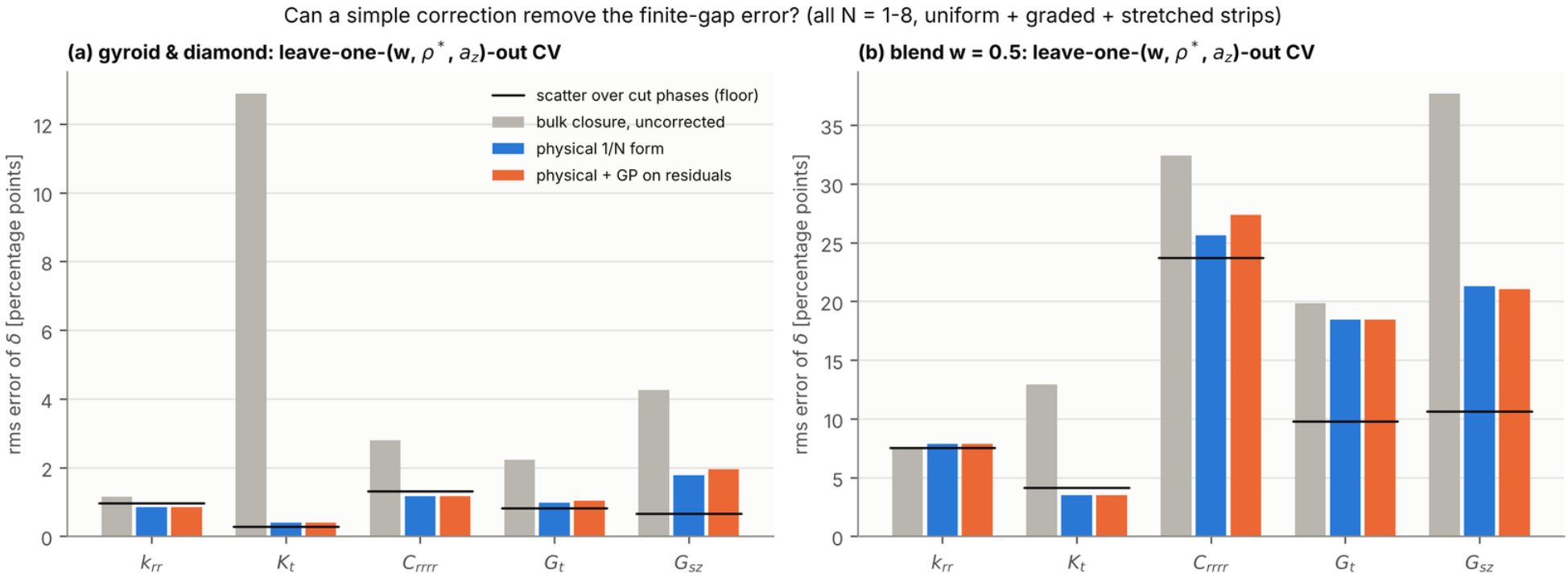
F19. Geometry-based (3D CNN on 32^3 voxels) vs parameter-based (GP, ensemble) learning of K* and E*: parity per split and error vs blend weight w.

File: results/figures/task6_cnn_extrapolation.png
Made by: scripts/task6_cnn_experiment.py (full run: ~80 min (2 cores))
Data: results/task6_cnn_metrics.csv



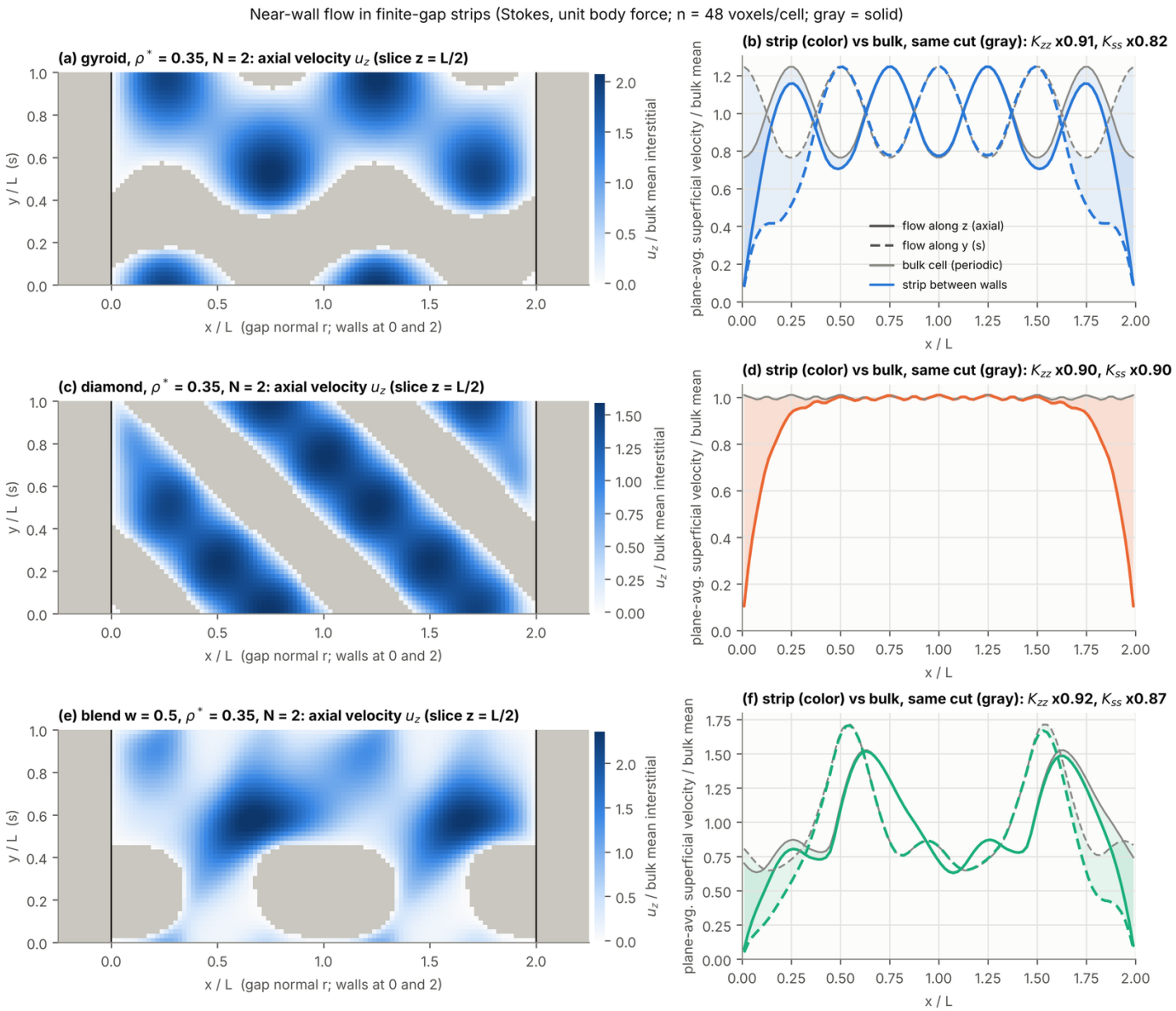
F20. Main RQ1 result: relative error delta of the bulk-homogenized prediction vs 1/N (N cells across a wall-bounded gap) for through-gap conductivity k_n, in-plane permeability K_t, normal stiffness C_nn and shear moduli G_t, G_sz; gyroid/diamond (top), blend (bottom); bars = range over cut phases, lines = physical correction delta = a/N; shaded band = jacket design range.

File: results/figures/task7_error_vs_N.png
Made by: scripts/task7_figures.py (full run: ~4 min (velocity slices n = 48))
Data: data/finite_gap.csv, results/task7_wall_coefficients.csv, results/task7_one_over_N.csv



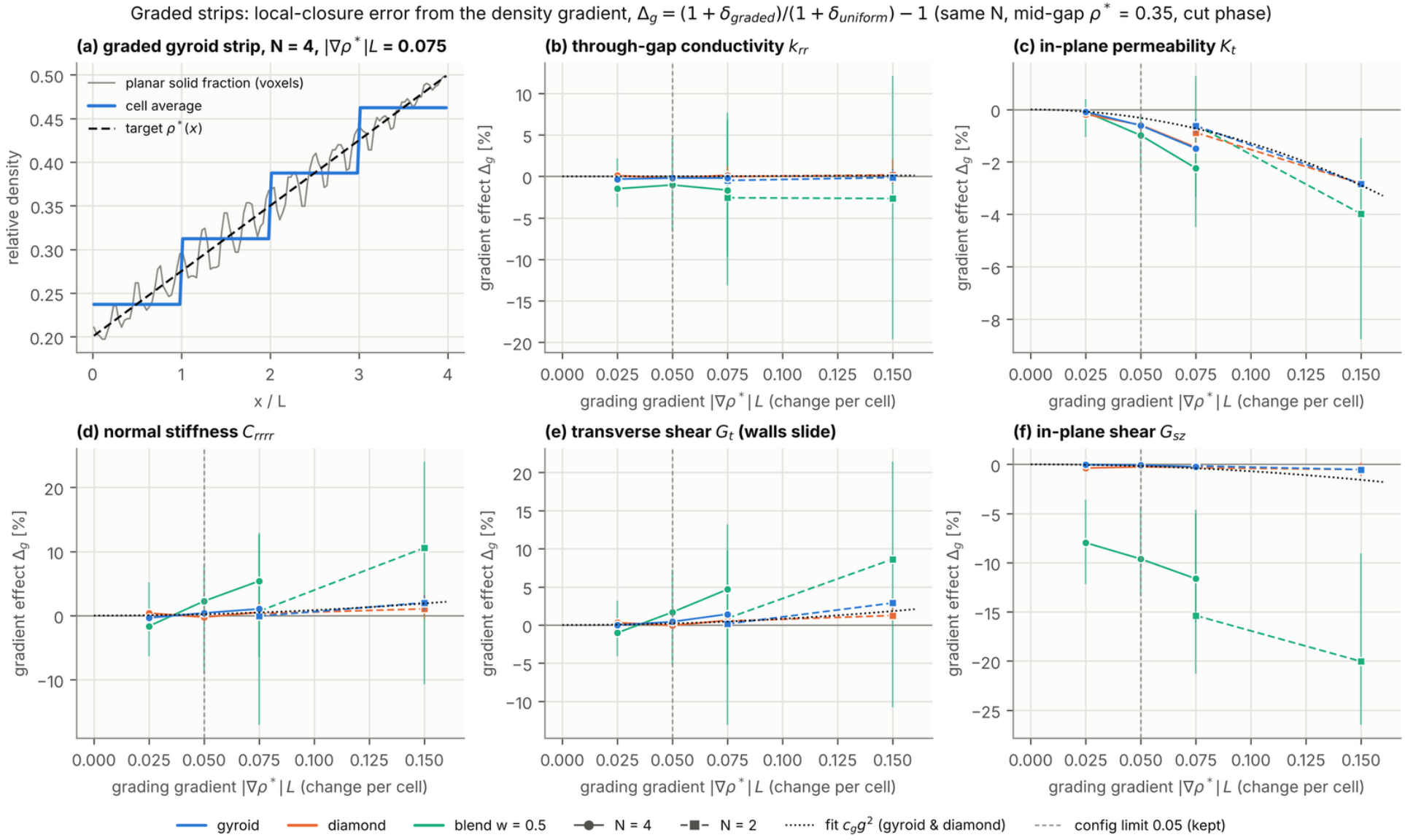
F21. Cross-validated error of the discrepancy models: uncorrected, physical a/N form, physical + GP on the residuals, and the cut-phase noise floor.

File: results/figures/task7_discrepancy_models.png
Made by: scripts/task7_figures.py (full run: ~4 min (velocity slices n = 48))
Data: results/task7_model_cv.csv



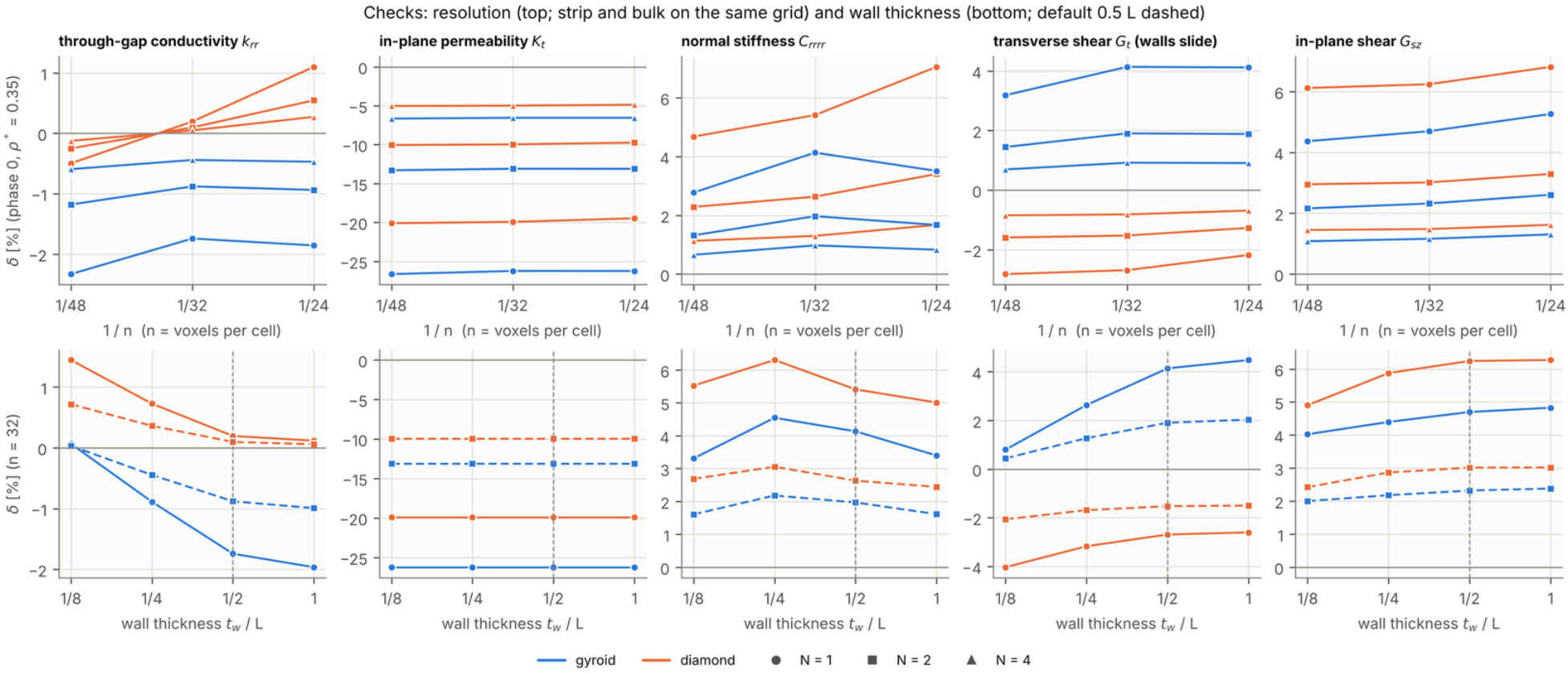
F22. Near-wall channelling: streamwise velocity slices and plane-averaged profiles in strips with N = 2 cells across the gap (n = 48).

File: results/figures/task7_velocity_slices.png
Made by: scripts/task7_figures.py (full run: ~4 min (velocity slices n = 48))



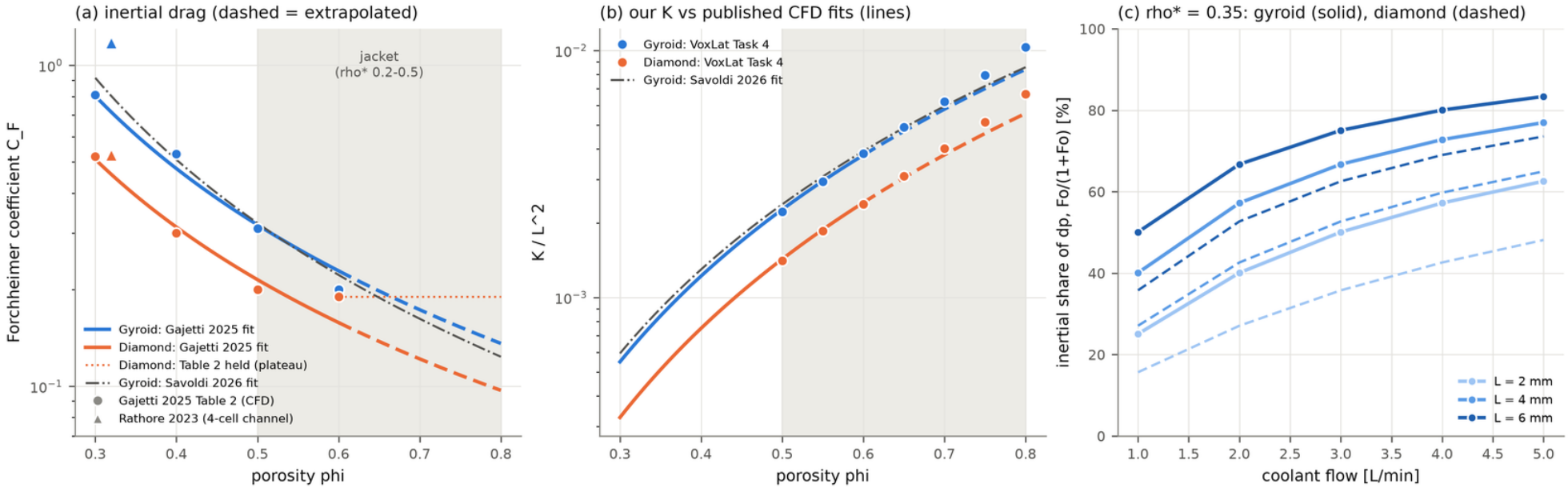
F23. Error of the homogenized prediction in strips with linear density grading vs the grading gradient |grad rho*| L.

File: results/figures/task7_graded_error.png
Made by: scripts/task7_figures.py (full run: ~4 min (velocity slices n = 48))
Data: results/task7_graded.csv



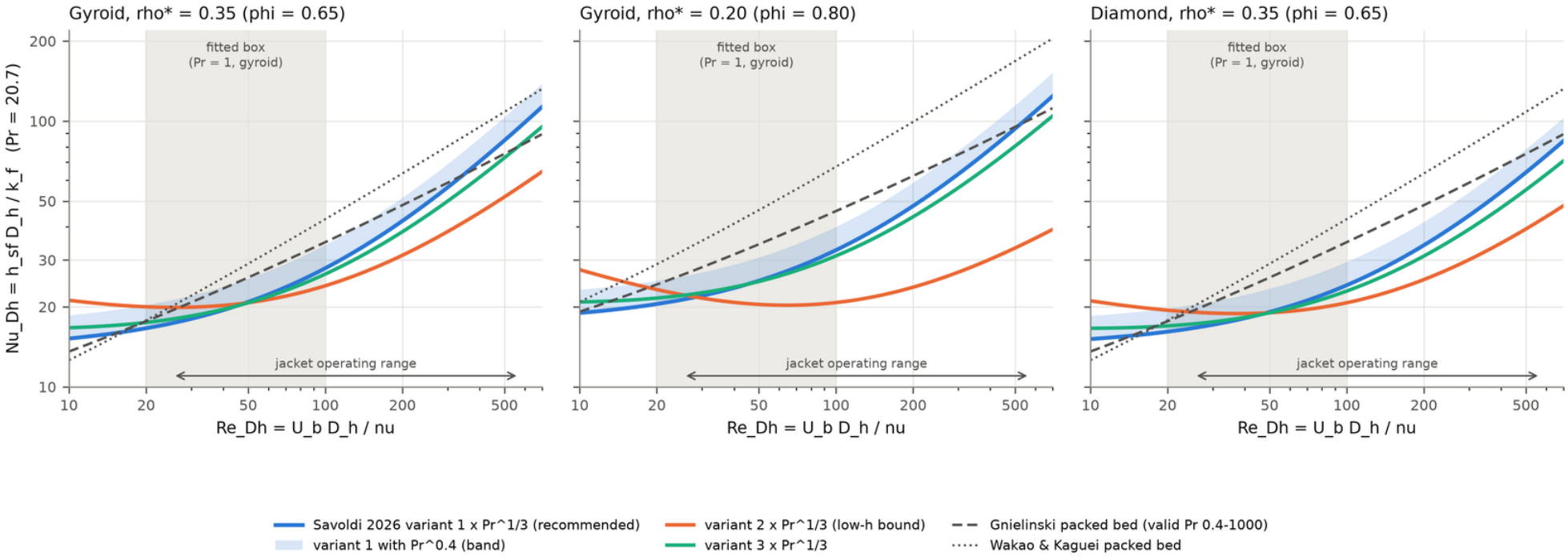
F24. Resolution and wall-thickness sensitivity of the finite-gap discrepancy.

File: results/figures/task7_convergence.png
Made by: scripts/task7_figures.py (full run: ~4 min (velocity slices n = 48))
Data: results/task7_convergence.csv, results/task7_wall_thickness.csv



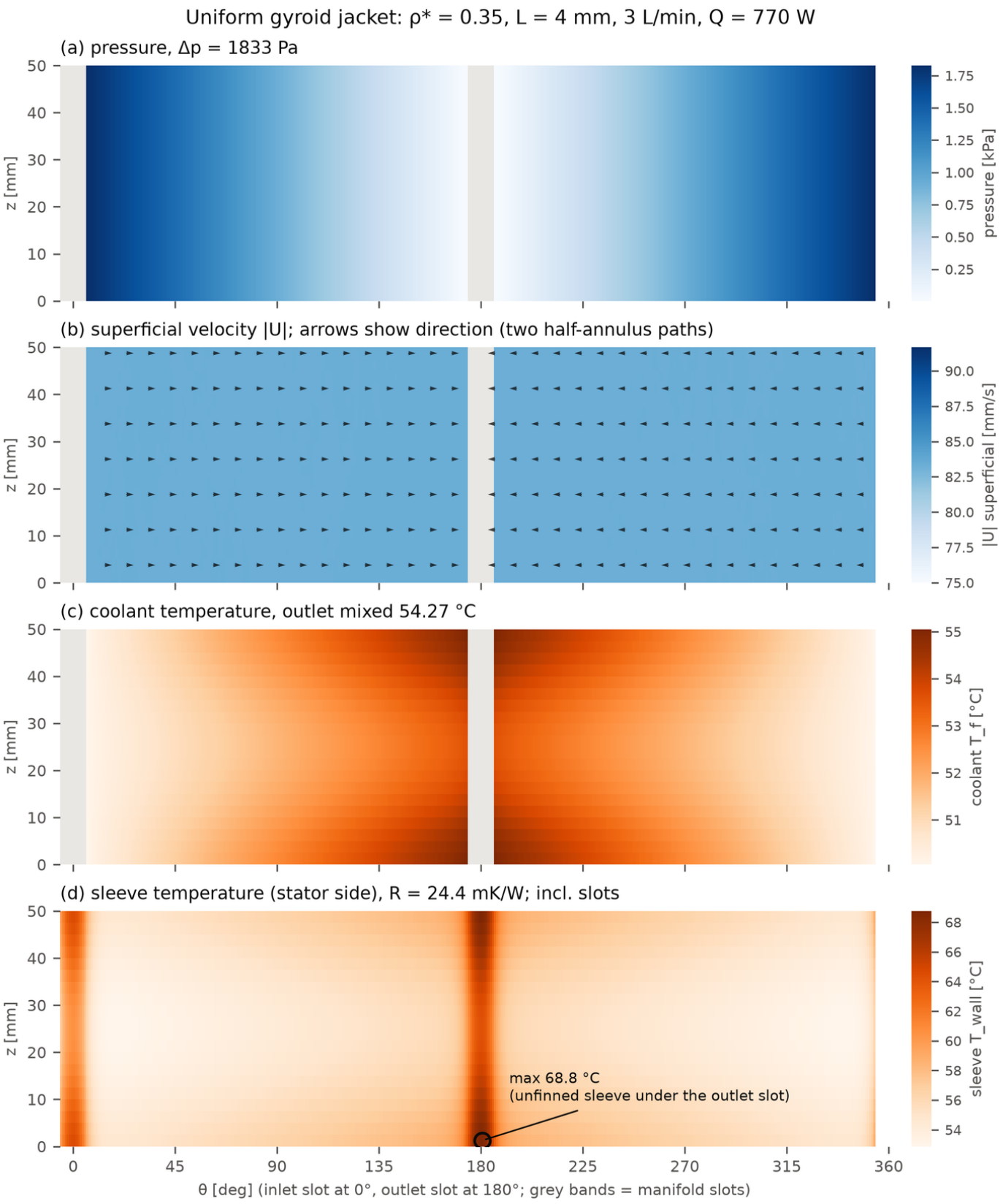
F25. Inertial closure: (a) Forchheimer coefficient C_F vs porosity (Gajetti et al. 2025 fits and points, Savoldi gyroid fit, Rathore points), jacket band; (b) Task 4 permeability vs the published fits; (c) inertial share of the pressure drop vs flow rate (rho* = 0.35, L = 2/4/6 mm).

File: results/figures/task8_forchheimer.png
Made by: scripts/task8_literature_closures.py (full run: 5 s)



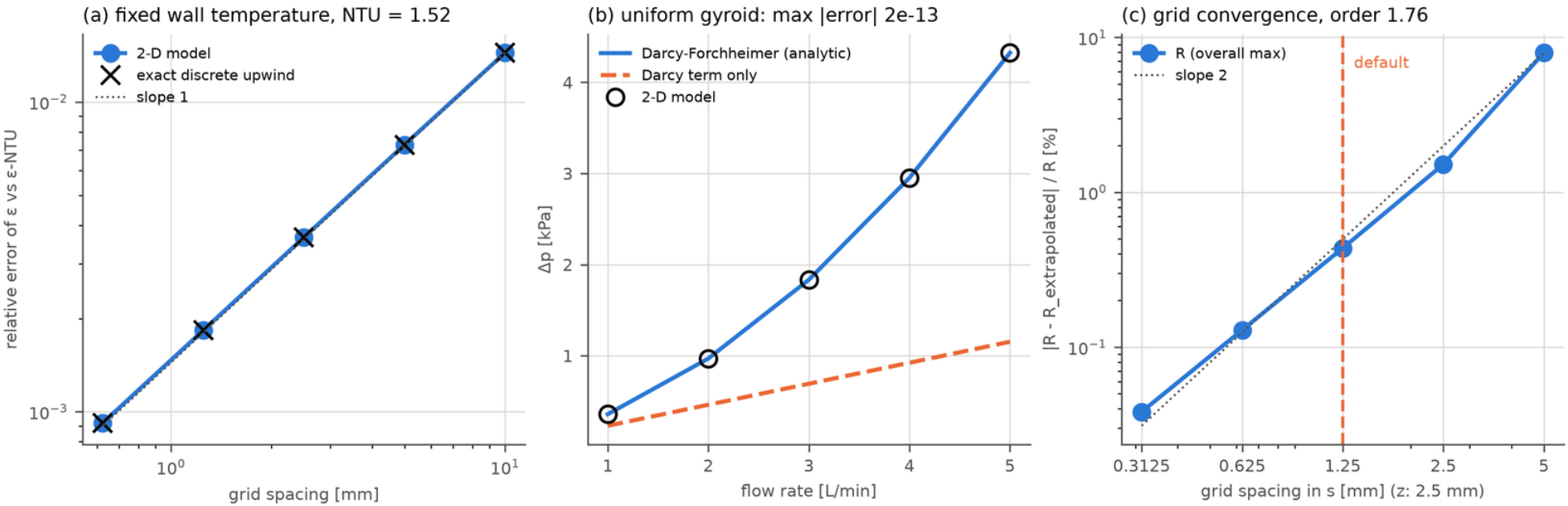
F26. Interstitial Nusselt number Nu_Dh vs Re_Dh at Pr = 20.7 for the candidate TPMS correlations (Pr^(1/3) and Pr^0.4 scalings) with Gnielinski and Wakao-Kaguei references; fitted box and jacket operating range.

File: results/figures/task8_nusselt.png
Made by: scripts/task8_literature_closures.py (full run: 5 s)
Data: results/task8_nusselt_comparison.csv, results/task8_operating_range.csv



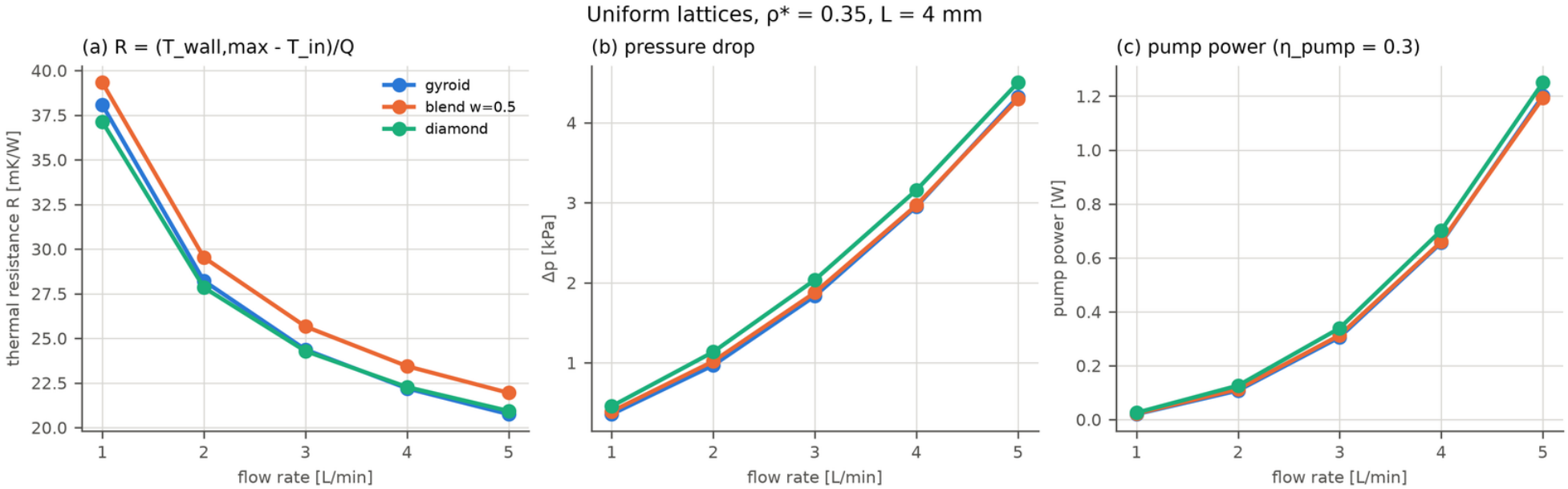
F27. Homogenized jacket model, uniform gyroid reference (rho* = 0.35, L = 4 mm, 3 L/min): pressure, Darcy velocity, coolant and stator-side sleeve temperature on the unwrapped (s, z) plane; slots shaded, hot spot marked.

File: results/figures/task9_fields_uniform_gyroid.png
Made by: scripts/task9_jacket_model.py (full run: ~20 s)
Data: results/task9_reference_gyroid.json



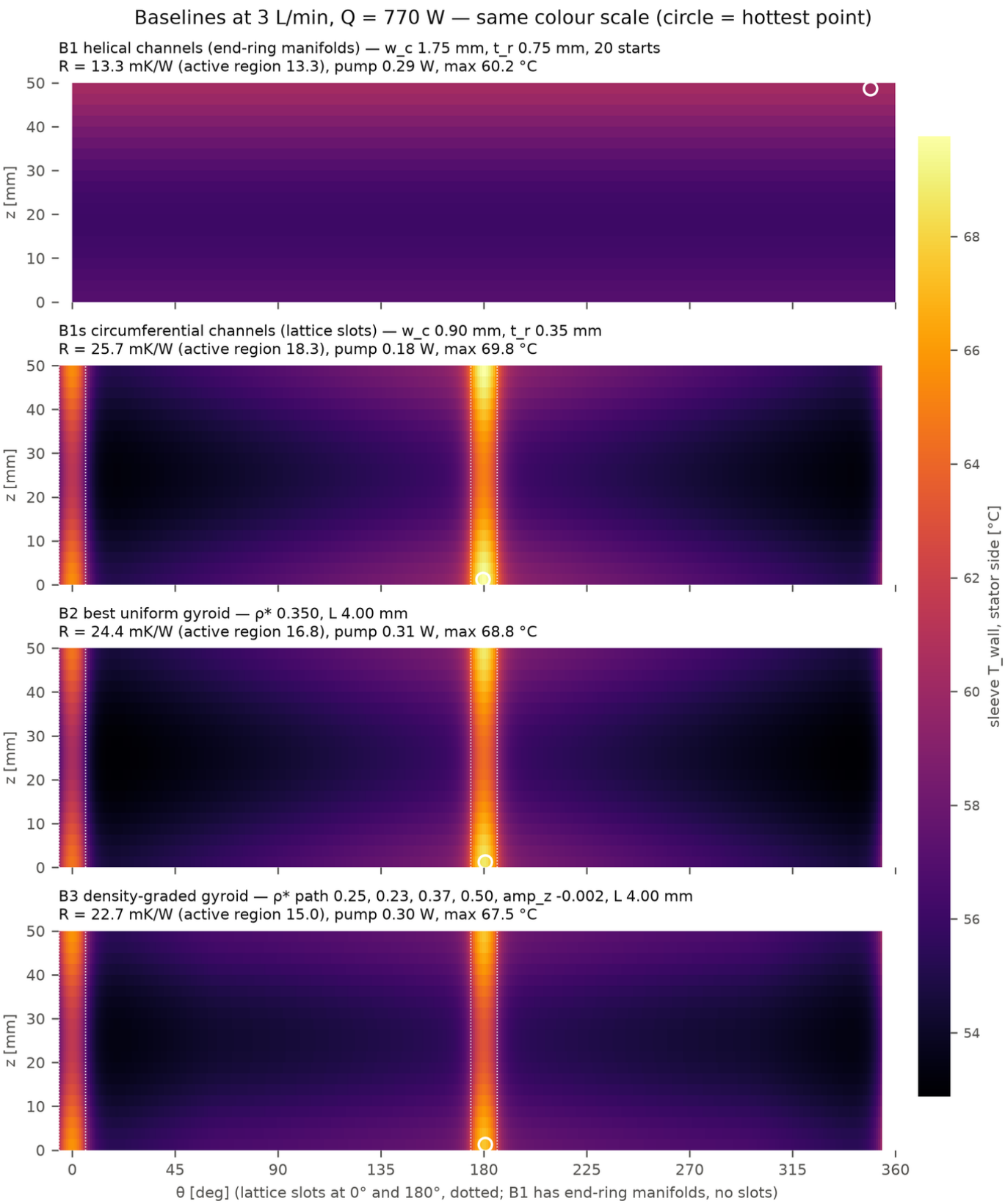
F28. Device-model verification: epsilon-NTU convergence against the exact discrete solution, pressure drop vs the analytic value over 1-5 L/min, and grid convergence of the thermal resistance R.

File: results/figures/task9_verification.png
Made by: scripts/task9_jacket_model.py (full run: ~20 s)
Data: results/task9_verification.csv, results/task9_grid_convergence.csv



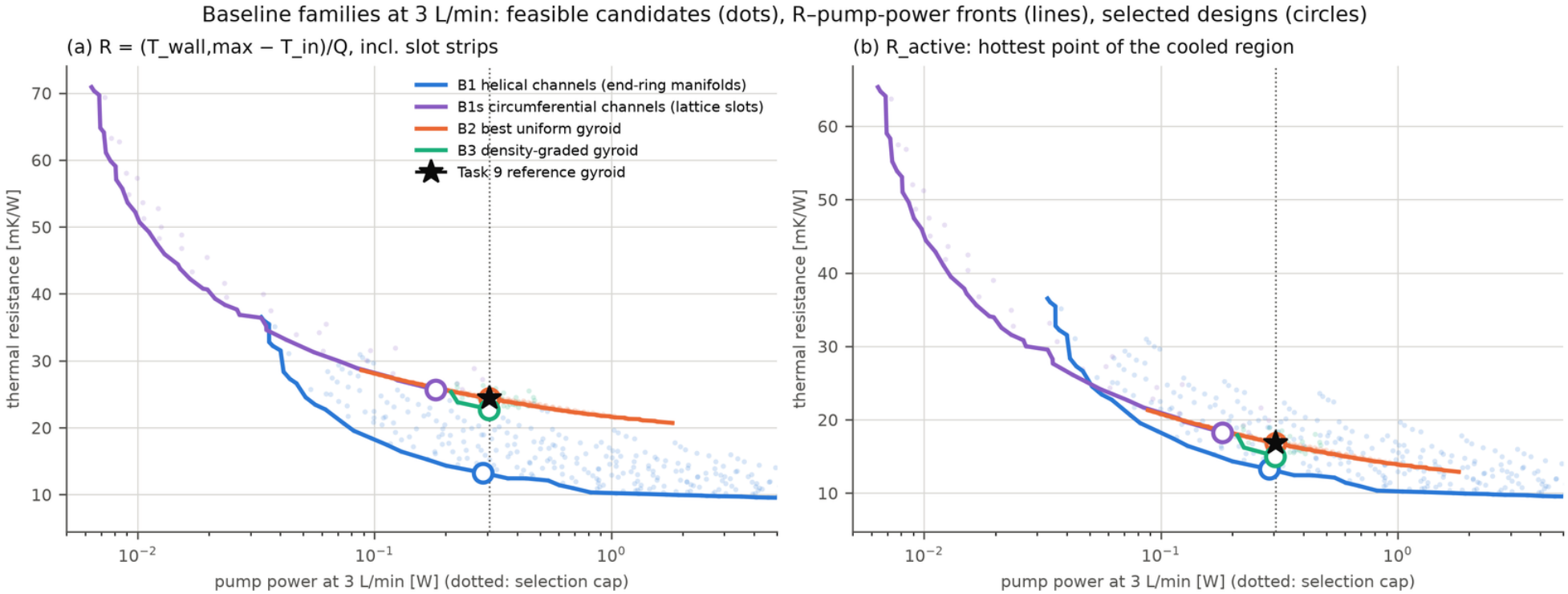
F29. Thermal resistance, pressure drop and pump power vs flow rate for uniform gyroid, blend and diamond.

File: results/figures/task9_flow_sweep.png
Made by: scripts/task9_jacket_model.py (full run: ~20 s)
Data: results/task9_flow_sweep.csv



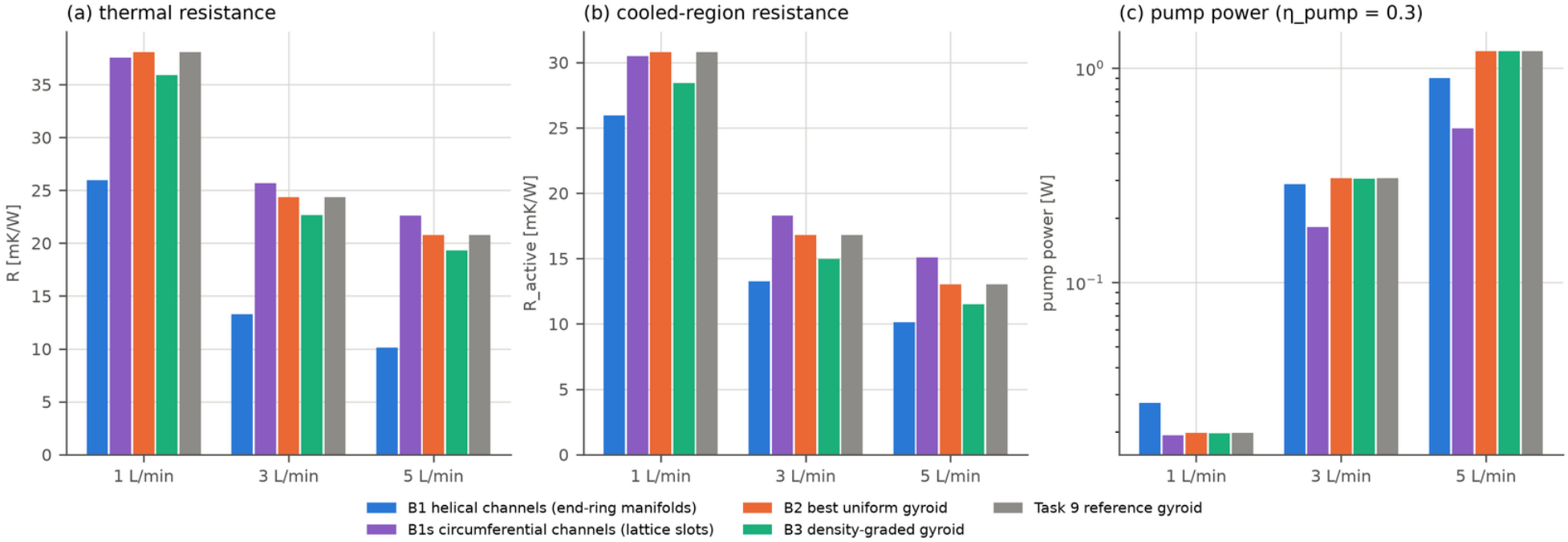
F30. Stator-side sleeve temperature of the baselines at 3 L/min on one colour scale: B1 helical channels (end-ring manifolds), B1s (slot manifolds), B2 best uniform gyroid, B3 density-only grading.

File: results/figures/task10_temperature_maps.png
Made by: scripts/task10_baselines.py (full run: ~2.5 min)



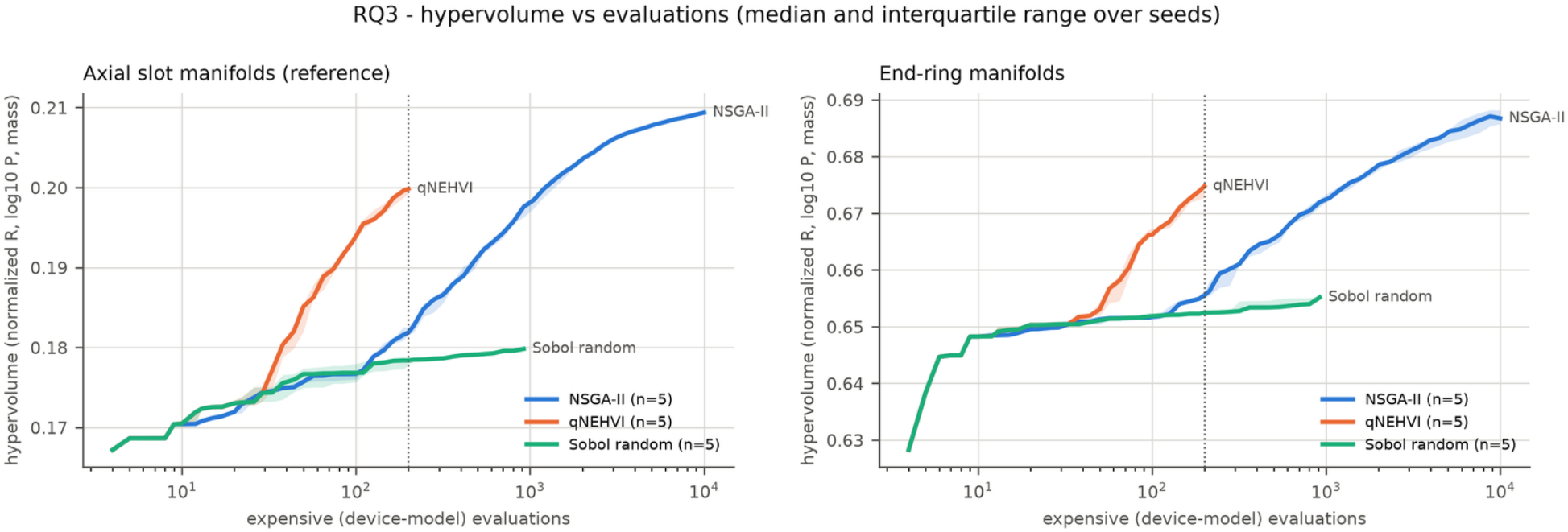
F31. Baseline tuning: R and R_active vs pump power for all candidates, the per-family fronts and the selected designs (pump power <= 0.306 W of the reference gyroid).

File: results/figures/task10_fronts.png
Made by: scripts/task10_baselines.py (full run: ~2.5 min)
Data: results/task10_fronts.csv



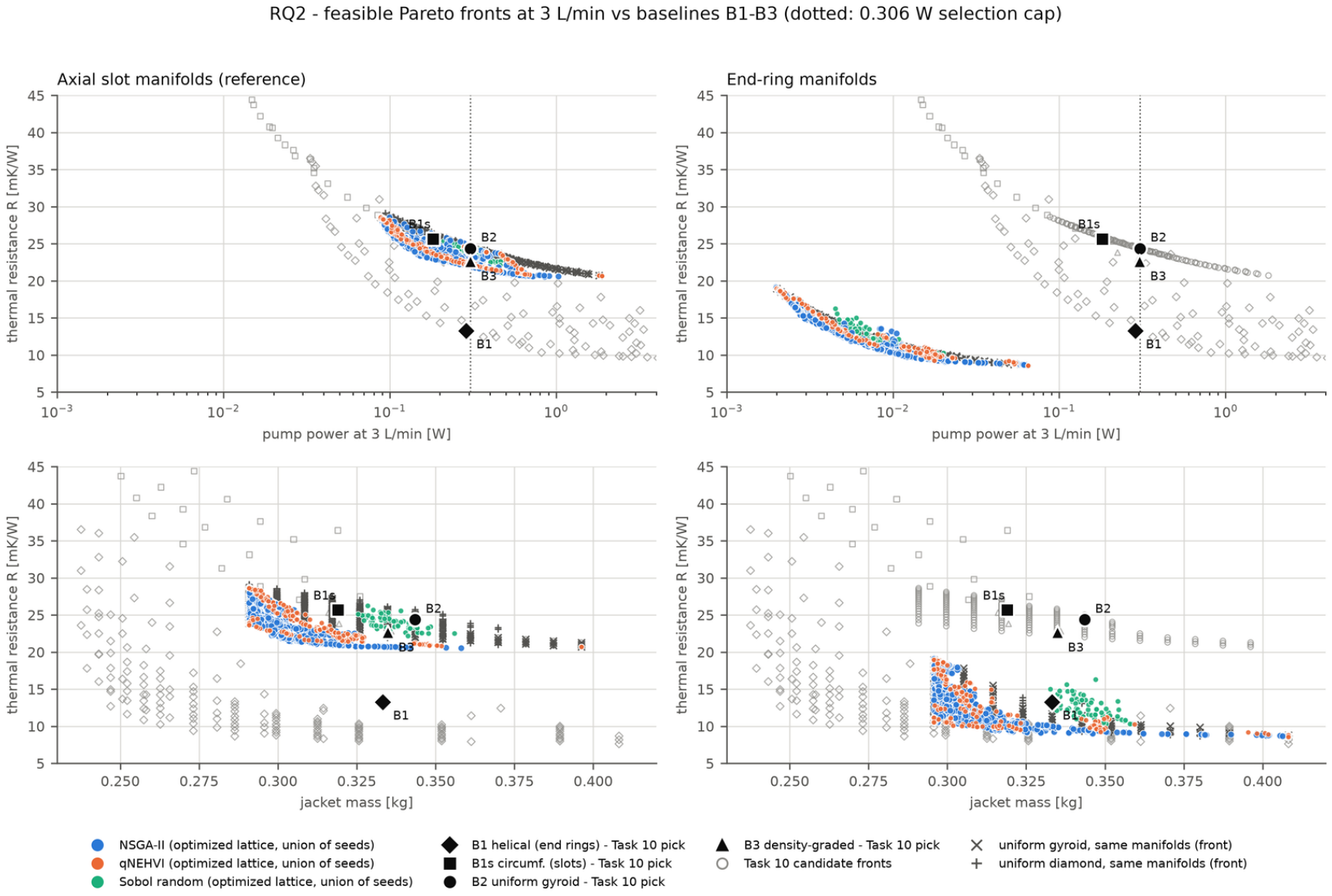
F32. Baselines B1-B3 at 1, 3 and 5 L/min: R, R_active and pump power.

File: results/figures/task10_flow_comparison.png
Made by: scripts/task10_baselines.py (full run: ~2.5 min)
Data: results/task10_comparison.csv



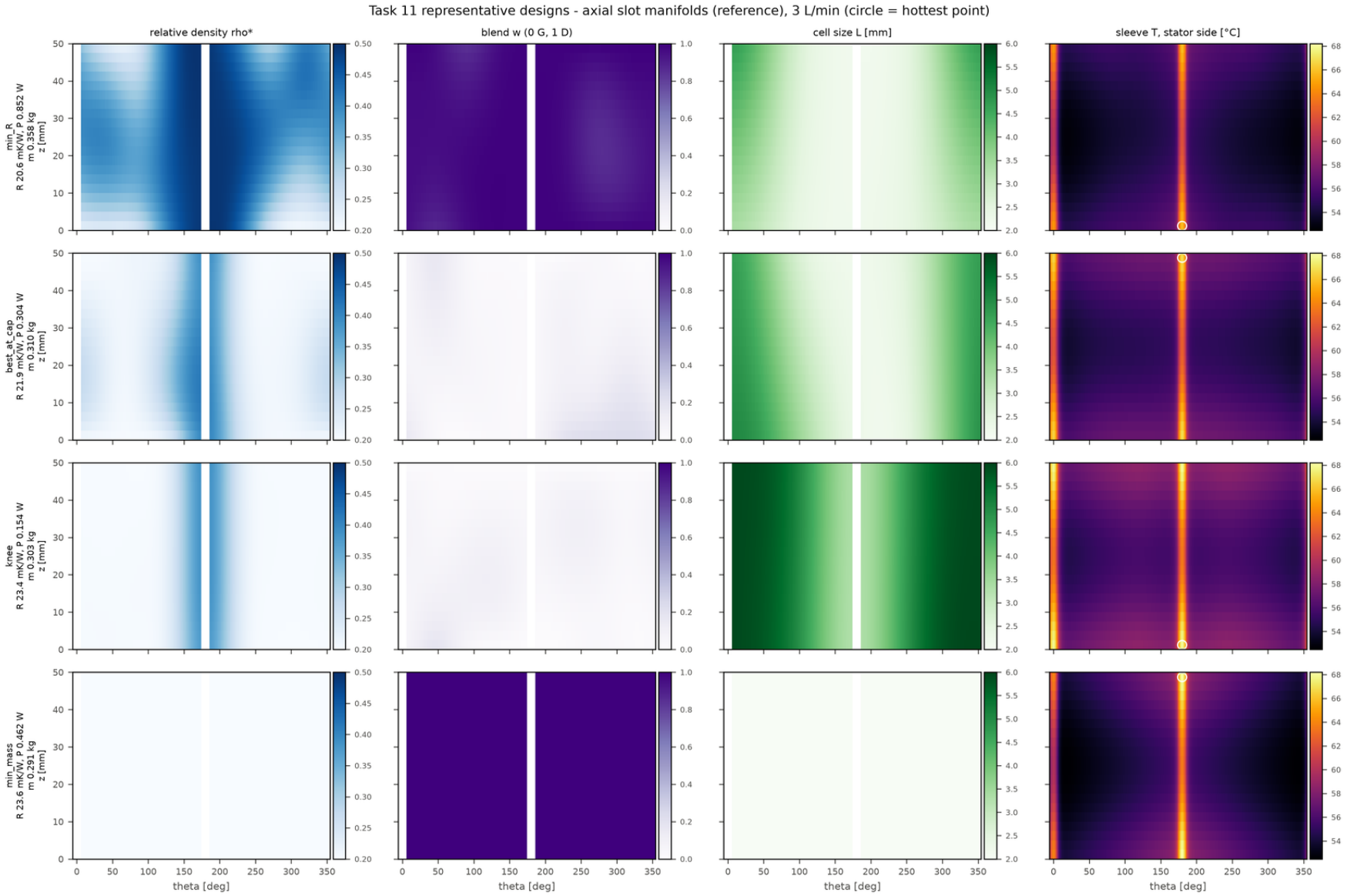
F33. Sample efficiency (RQ3): normalized hypervolume vs evaluations for NSGA-II, qNEHVI and Sobol search, median and IQR over 5 seeds, slot and end-ring manifolds.

File: results/figures/task11_hypervolume.png
Made by: scripts/task11_analysis.py (full run: ~5 min)
Data: results/task11_hv_curves.csv, results/task11_hv_summary.csv



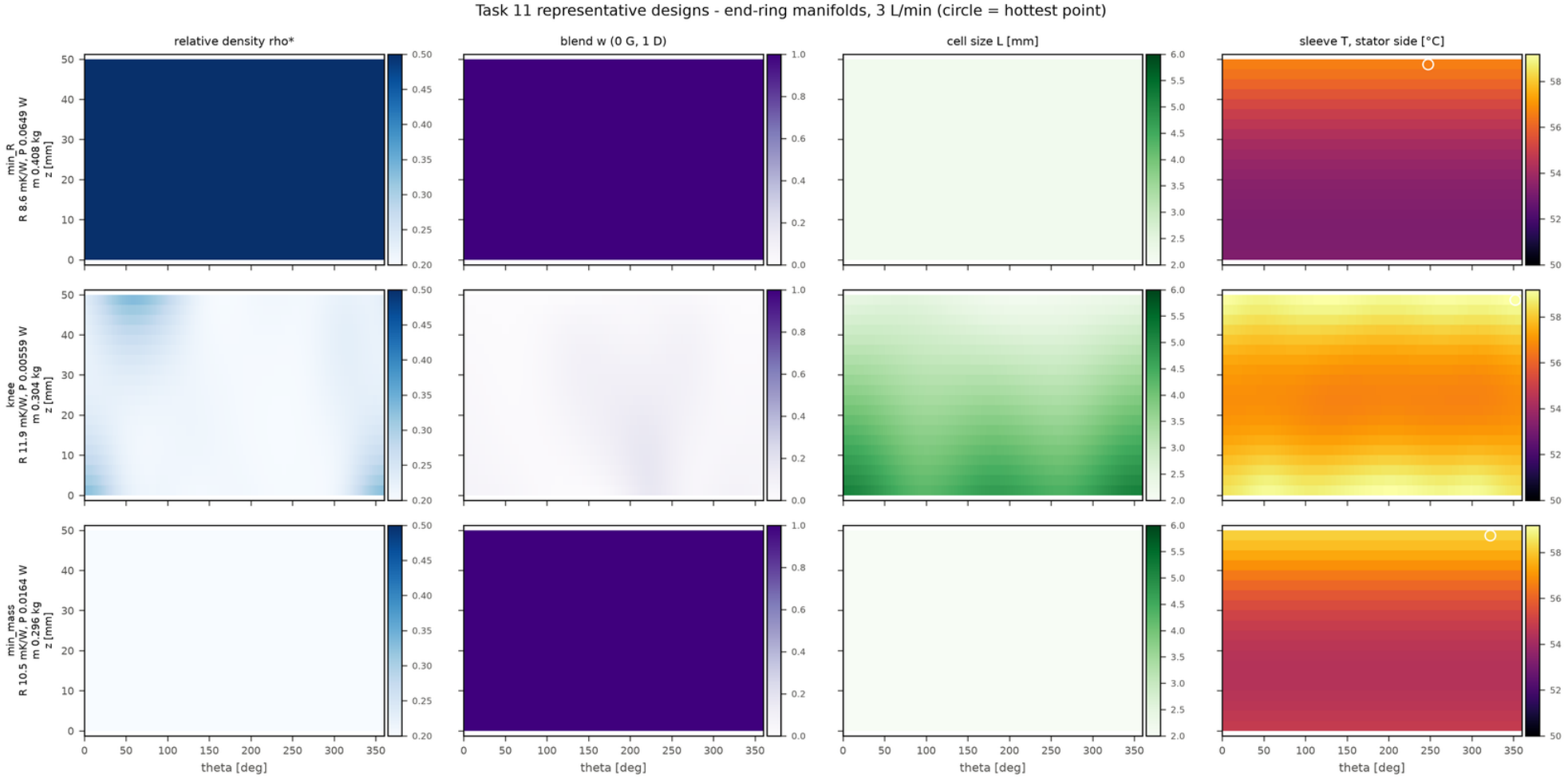
F34. Pareto fronts (R, pump power, mass at 3 L/min) of the graded lattices on both manifold layouts with baselines B1-B3 and uniform gyroid/diamond references overlaid.

File: results/figures/task11_pareto_fronts.png
Made by: scripts/task11_analysis.py (full run: ~5 min)
Data: results/task11_front_slots.csv, results/task11_front_end_rings.csv



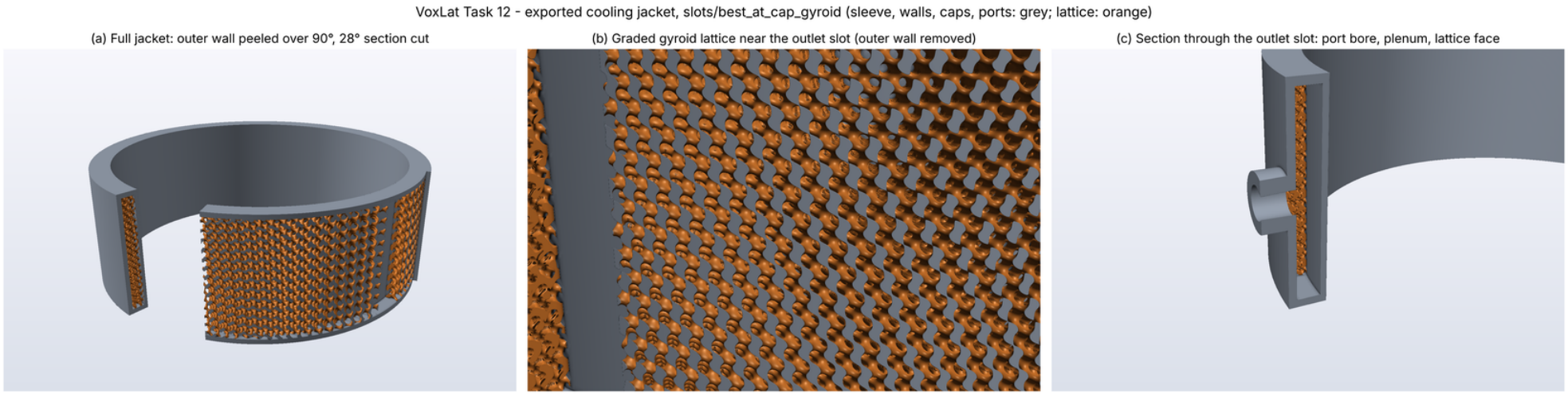
F35. Representative slot-manifold designs (min-R, best at the 0.306 W cap, knee, min-mass): rho*, w and L fields on the unwrapped jacket. The best-at-cap design is exported with w = 0 (task12_slots_best_at_cap_gyroid) as the manufacturable optimum.

File: results/figures/task11_designs_slots.png
Made by: scripts/task11_analysis.py (full run: ~5 min)



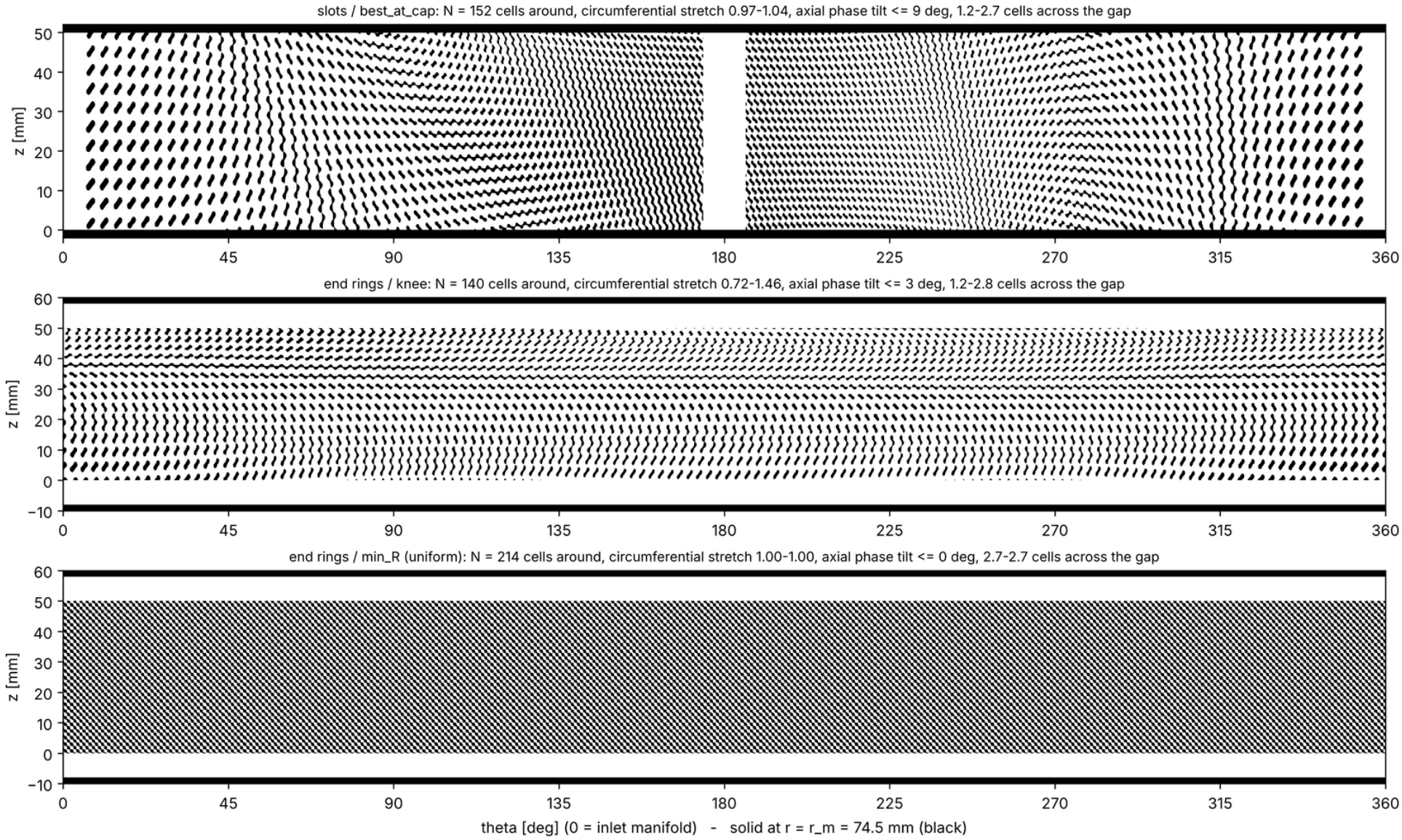
F36. Representative end-ring-manifold designs (min-R, knee, min-mass): rho*, w and L fields.

File: results/figures/task11_designs_end_rings.png
Made by: scripts/task11_analysis.py (full run: ~5 min)



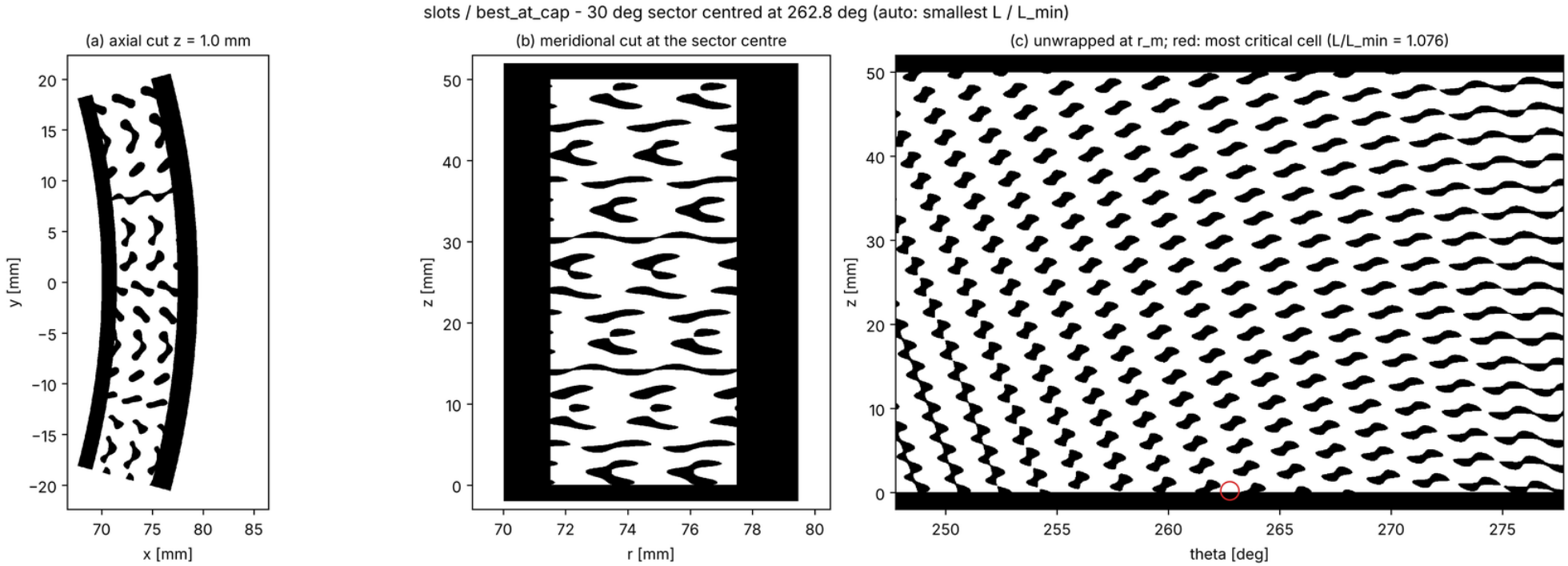
F37. The exported manufacturable optimum (task12_slots_best_at_cap_gyroid), ray-traced from the signed-distance field that the PicoGK export reproduces voxel by voxel (IoU 0.997): overview, lattice close-up and outlet-slot section.

File: results/figures/task12_renders.png
Made by: scripts/task12_render.py (full run: ~2 min at 1800 px)
Data: results/figures/task12_render_overview.png, results/figures/task12_render_lattice_closeup.png, results/figures/task12_render_outlet_slot_section.png



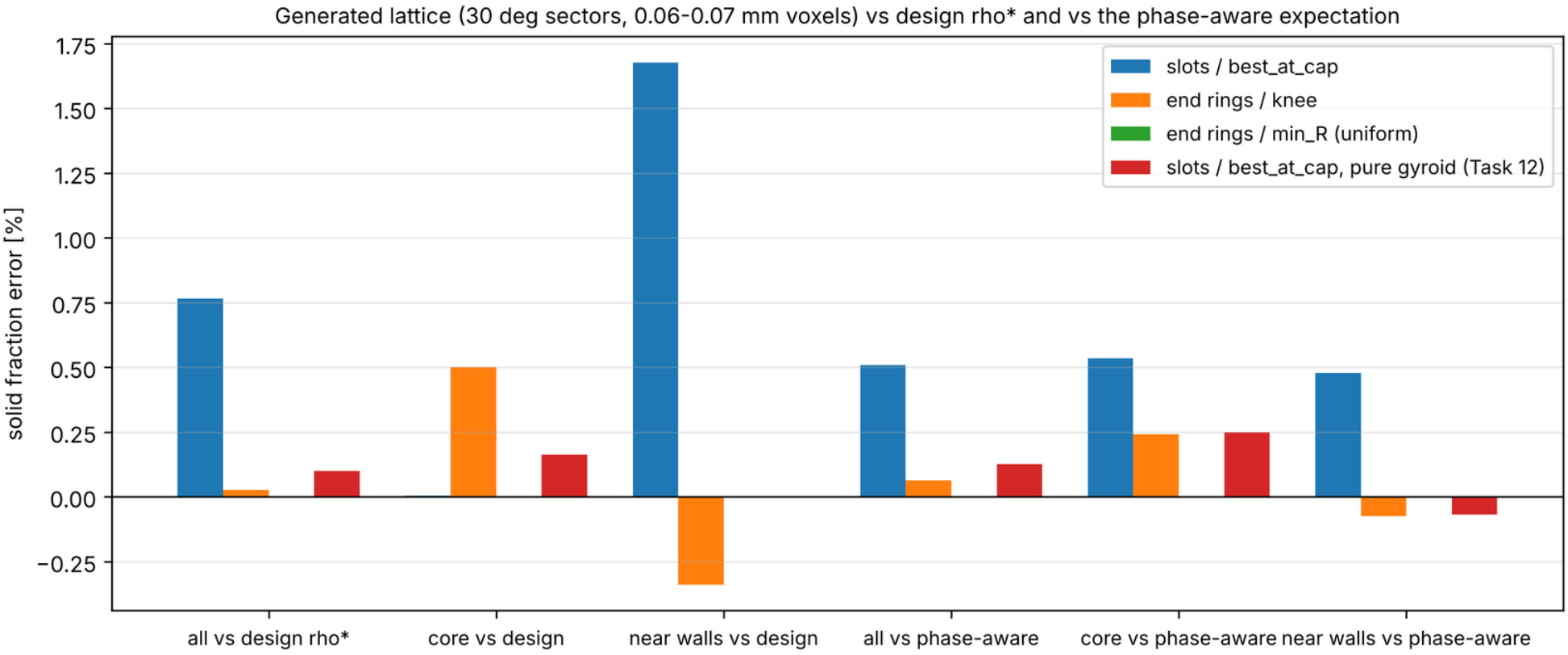
F38. Exported lattice unwrapped at the mean gap radius for three Task 11 designs: conformal phase-field grading with an integer number of cells around, no torn cells.

File: results/figures/task12_phase_field.png
Made by: scripts/task12_figures.py (full run: ~3 min)



F39. 30-degree sector of slots/best_at_cap around its most critical cell: axial, meridional and unwrapped cuts with sleeve, outer wall, end caps, slot and port (the blend version that fails the 0.35 mm wall check).

File: results/figures/task12_sector_slices.png
Made by: scripts/task12_figures.py (full run: ~3 min)



F40. Volume fraction of the exported sectors vs the design rho* and the phase-aware expectation, overall, core / near-wall and per patch.

File: results/figures/task12_volume_fraction.png
Made by: scripts/task12_figures.py (full run: ~3 min)

T01. Production estimators of the cell solvers: convergence order, raw and R(32, 64) errors against the extrapolated reference (k_eff, C_eff, K) for gyroid and diamond.

quantity	w	rho	case	p_observed	err_raw_n32	err_raw_n64	err_R32_64
k_eff/k_s	0	0.2	k_f/k_s=0.003077	1.007	-0.1151	-0.05457	0.00599
k_eff/k_s	0	0.35	k_f/k_s=0	0.7704	-0.08584	-0.04055	0.004731
k_eff/k_s	0	0.35	k_f/k_s=0.003077	0.772	-0.08332	-0.03937	0.004585
k_eff/k_s	0	0.5	k_f/k_s=0.003077	0.9108	-0.05788	-0.03003	-0.002169
k_eff/k_s	1	0.2	k_f/k_s=0.003077	0.9366	-0.1737	-0.0834	0.006924
k_eff/k_s	1	0.35	k_f/k_s=0	1.063	-0.112	-0.05362	0.004794
k_eff/k_s	1	0.35	k_f/k_s=0.003077	1.063	-0.1087	-0.052	0.004659
k_eff/k_s	1	0.5	k_f/k_s=0.003077	0.9875	-0.08071	-0.03989	0.0009269
C_eff E_axial	0	0.2		1.506	-0.06902	-0.02789	0.01323
C_eff C11c	0	0.2		1.57	-0.04863	-0.02122	0.006198
C_eff C12c	0	0.2		1.661	-0.03369	-0.01633	0.001018
C_eff C44c	0	0.2		0.5766	-0.04702	-0.02072	0.005577
C_eff E_axial	0	0.35		1.966	-0.04481	-0.009474	0.02586
C_eff C11c	0	0.35		1.592	-0.03549	-0.01041	0.01468
C_eff C12c	0	0.35		1.174	-0.02269	-0.01175	-0.0008113
C_eff C44c	0	0.35		0.25	-0.02097	-0.01043	0.0001105
C_eff E_axial	0	0.5		0.25	-0.009476	-0.005104	-0.0007312
C_eff C11c	0	0.5		0.6902	-0.01253	-0.006988	-0.001447
C_eff C12c	0	0.5		1.471	-0.01829	-0.01055	-0.002802
C_eff C44c	0	0.5		1.498	-0.01512	-0.008009	-0.0009003
C_eff E_axial	1	0.2		1.348	-0.04139	-0.01132	0.01875
C_eff C11c	1	0.2		1.151	-0.06344	-0.02053	0.02237
C_eff C12c	1	0.2		1.282	-0.08436	-0.02986	0.02465
C_eff C44c	1	0.2		0.6124	-0.04266	-0.01354	0.01559
C_eff E_axial	1	0.35		0.25	0.007651	-0.00101	-0.009672
C_eff C11c	1	0.35		0.25	-0.01708	-0.008751	-0.0004197
C_eff C12c	1	0.35		1.74	-0.04701	-0.01829	0.01044
C_eff C44c	1	0.35		0.25	-0.005096	-0.003978	-0.002859
C_eff E_axial	1	0.5		0.25	-0.0006812	-0.002942	-0.005202
C_eff C11c	1	0.5		0.25	-0.01384	-0.006966	-8.879e-05
C_eff C12c	1	0.5		1.188	-0.03257	-0.01272	0.007135
C_eff C44c	1	0.5		0.25	-0.004096	-0.002836	-0.001576
K/L^2	0	0.2		4	0.0005815	0.001078	0.001575
K/L^2	0	0.35		0.25	-0.00806	-0.003274	0.001511

Full table: results/tables/T01_solver_verification.csv (+ .md). Made by scripts/experiments/tables.py T01

T01. Production estimators of the cell solvers: convergence order, raw and R(32, 64) errors against the extrapolated reference (k_eff, C_eff, K) for gyroid and diamond.

quantity	w	rho	case	p_observed	err_raw_n32	err_raw_n64	err_R32_64
K/L^2	0	0.5		1.601	-0.002654	-0.001939	-0.001223
K/L^2	0.5	0.35		4	-0.00894	-0.00389	0.001159
K/L^2	1	0.2		4	-0.003759	-0.003308	-0.002857
K/L^2	1	0.35		0.25	-0.009089	-0.002807	0.003475
K/L^2	1	0.5		0.25	-0.03237	-0.01107	0.01022

Full table: results/tables/T01_solver_verification.csv (+ .md). Made by scripts/experiments/tables.py T01

T02. Closure surrogate accuracy (R^2, MAPE, 95 % coverage) for the GP and deep ensemble on the cv5, band and sparse splits.

model	split	group	R2_median	MAPE_median	cov95_median
gp+cal	cv5	K principal	0.999	0.008858	0.9138
gp+cal	cv5	k_eff	0.9951	0.01905	0.9552
gp+cal	cv5	C_eff	0.8801	0.06345	0.9621
gp+cal	cv5	E	0.9884	0.04709	0.9517
gp+cal	cv5	a_sf L	0.999	0.002561	0.9483
gp+cal	cv5	localization p99	0.9853	0.04787	0.9414
gp+cal	band	K principal	0.9731	0.06478	0.3485
gp+cal	band	k_eff	0.9195	0.1145	0.8561
gp+cal	band	C_eff	-0.1737	0.4716	0.6818
gp+cal	band	E	0.8317	0.4282	0.7576
gp+cal	band	a_sf L	0.9453	0.02272	0.4924
gp+cal	band	localization p99	0.8687	0.1598	0.7197
gp+cal	sparse	K principal	0.9955	0.02112	0.8974
gp+cal	sparse	k_eff	0.982	0.03743	0.9487
gp+cal	sparse	C_eff	0.5088	0.1062	0.9692
gp+cal	sparse	E	0.9667	0.09552	0.9538
gp+cal	sparse	a_sf L	0.9956	0.005323	0.8718
gp+cal	sparse	localization p99	0.9749	0.06862	0.959
ensemble+cal	cv5	K principal	0.9988	0.00907	0.9069
ensemble+cal	cv5	k_eff	0.9954	0.01452	0.9448
ensemble+cal	cv5	C_eff	0.9576	0.03849	0.9517
ensemble+cal	cv5	E	0.9888	0.04055	0.9345
ensemble+cal	cv5	a_sf L	0.9992	0.002313	0.9793
ensemble+cal	cv5	localization p99	0.9887	0.03338	0.9328
ensemble+cal	band	K principal	0.981	0.0524	0.8409
ensemble+cal	band	k_eff	0.9158	0.1213	0.8106
ensemble+cal	band	C_eff	0.2037	0.517	0.7197
ensemble+cal	band	E	0.805	0.4991	0.7652
ensemble+cal	band	a_sf L	0.9366	0.02374	0.8636
ensemble+cal	band	localization p99	0.8523	0.1754	0.7008
ensemble+cal	sparse	K principal	0.9936	0.0241	0.8103
ensemble+cal	sparse	k_eff	0.9827	0.03332	0.9333
ensemble+cal	sparse	C_eff	0.799	0.1036	0.9179
ensemble+cal	sparse	E	0.9623	0.09918	0.8974

Full table: results/tables/T02_surrogate_accuracy.csv (+ .md). Made by scripts/experiments/tables.py T02

T02. Closure surrogate accuracy (R^2, MAPE, 95 % coverage) for the GP and deep ensemble on the cv5, band and sparse splits.

model	split	group	R2_median	MAPE_median	cov95_median
ensemble+cal	sparse	a_sf L	0.9941	0.006283	0.8667
ensemble+cal	sparse	localization p99	0.9736	0.06235	0.9051

T05. Baselines B1, B1s, B2, B3 at 1, 3 and 5 L/min: R, R_active, pressure drop, pump power, mass, margin.

family	flow_lpm	R_mK_W	R_active_mK_W	pump_power_W	mass_kg	min_margin
B1	1	25.94	25.94	0.02733	0.3332	9.255
B1	3	13.26	13.26	0.2867	0.3332	9.244
B1	5	10.12	10.12	0.8968	0.3332	9.231
B1s	3	25.67	18.27	0.1814	0.319	4.246
B2	3	24.37	16.81	0.3055	0.3435	12.69
B3	3	22.67	14.97	0.3045	0.3349	12.28

T06. Hypervolume (median, IQR over 5 seeds) at 50 / 100 / 200 evaluations and final budget; evaluations NSGA-II needs to match qNEHVI at 200.

layout	method	hv_50_median	hv_100_median	hv_200_median	n_evals_final_median	hv_final_median	evals_to_reach_qnehvi_hv200_median
end_rings	nsga2	0.6513	0.6517	0.6555	10000	0.6875	1368
end_rings	qnehvi	0.653	0.6662	0.6748	200	0.6748	200
end_rings	sobol	0.6511	0.6519	0.6525	919	0.6552	
slots	nsga2	0.1757	0.1767	0.1819	10000	0.2093	1368
slots	qnehvi	0.1851	0.1939	0.1998	200	0.1998	200
slots	sobol	0.1767	0.1768	0.1784	919	0.1798	

T07. RQ2 at equal pump power (3 L/min): optimized R vs the best baseline of each family below the pump-power cap; slots row at the 0.306 W cap uses the manufacturable optimum task12_slots_best_at_cap_gyroid.

layout	design	cap_W	family	R_baseline_mK_W	P_baseline_W	R_opt_mK_W	P_opt_W	R_change_pct
slots	Task 11 optimum (archive)	0.3055	B1s	25.67	0.1814	21.94	0.3039	-14.52
slots	Task 11 optimum (archive)	0.3055	B2	24.46	0.2931	21.94	0.3039	-10.31
slots	Task 11 optimum (archive)	0.3055	B3	22.67	0.3045	21.94	0.3039	-3.221
slots	Task 11 optimum (archive)	0.3055	U-G	24.46	0.2931	21.94	0.3039	-10.31
slots	Task 11 optimum (archive)	0.3055	U-D	24.65	0.2993	21.94	0.3039	-10.99
slots	Task 11 optimum (archive)	0.3055	B1	13.26	0.2867	21.94	0.3039	65.46
end_rings	Task 11 optimum (archive)	0.3055	B1	13.26	0.2867	8.573	0.06487	-35.35
end_rings	Task 11 optimum (archive)	0.3055	U-G	8.811	0.05359	8.573	0.06487	-2.707
end_rings	Task 11 optimum (archive)	0.3055	U-D	8.634	0.06158	8.573	0.06487	-0.7085
end_rings	Task 11 optimum (archive)	0.3055	B1s	25.67	0.1814	8.573	0.06487	-66.6
end_rings	Task 11 optimum (archive)	0.3055	B2	24.46	0.2931	8.573	0.06487	-64.96
end_rings	Task 11 optimum (archive)	0.3055	B3	22.67	0.3045	8.573	0.06487	-62.19
slots	best_at_cap_gyroid, 0.306 W cap (Task 11 rule)	0.3055	B1s	25.67	0.1814	21.92	0.3076	-14.59
slots	best_at_cap_gyroid, 0.306 W cap (Task 11 rule)	0.3055	B2	24.46	0.2931	21.92	0.3076	-10.39
slots	best_at_cap_gyroid, 0.306 W cap (Task 11 rule)	0.3055	B3	22.67	0.3045	21.92	0.3076	-3.308
slots	best_at_cap_gyroid, 0.306 W cap (Task 11 rule)	0.3055	U-G	24.46	0.2931	21.92	0.3076	-10.39
slots	best_at_cap_gyroid, 0.306 W cap (Task 11 rule)	0.3055	U-D	24.65	0.2993	21.92	0.3076	-11.07
slots	best_at_cap_gyroid, 0.306 W cap (Task 11 rule)	0.3055	B1	13.26	0.2867	21.92	0.3076	65.32
slots	best_at_cap_gyroid, matched: cap = P of the gyro	0.3076	B1s	25.67	0.1814	21.92	0.3076	-14.59
slots	best_at_cap_gyroid, matched: cap = P of the gyro	0.3076	B2	24.37	0.3055	21.92	0.3076	-10.04
slots	best_at_cap_gyroid, matched: cap = P of the gyro	0.3076	B3	22.67	0.3045	21.92	0.3076	-3.308
slots	best_at_cap_gyroid, matched: cap = P of the gyro	0.3076	U-G	24.37	0.3055	21.92	0.3076	-10.04
slots	best_at_cap_gyroid, matched: cap = P of the gyro	0.3076	U-D	24.65	0.2993	21.92	0.3076	-11.07
slots	best_at_cap_gyroid, matched: cap = P of the gyro	0.3076	B1	13.26	0.2867	21.92	0.3076	65.32

T08. Representative designs per layout (R, R_active, pump power, mass, margin, field ranges), including the manufacturable slot optimum best_at_cap_gyroid.

layout	design	R_mK_W	R_active_mK_W	pump_power_W	mass_kg	min_margin	rho_min	rho_max	w_max	L_min_mm	L_max_mm
slots	min_R	20.57	12.94	0.8515	0.3581	12.49	0.2113	0.496	0.9969	2.183	4.671
slots	best_at_cap	21.94	14.37	0.3039	0.3096	8.743	0.2014	0.4149	0.2354	2.188	5.008
slots	knee	23.36	15.76	0.1541	0.3026	8.904	0.2007	0.3873	0.1777	3.303	5.961
slots	min_mass	23.64	16.09	0.4623	0.2909	12.63	0.2	0.2	1	2	2
end_rings	min_R	8.573	8.573	0.06487	0.4082	47.65	0.5	0.5	1	2.187	2.187
end_rings	knee	11.93	11.93	0.005589	0.304	9.011	0.2017	0.3303	0.1689	2.179	5.141
end_rings	min_mass	10.54	10.54	0.01641	0.2957	13.32	0.2	0.2	1	2	2
slots	best_at_cap_gyroid (manufacturable, exported)	21.92	14.41	0.3076	0.3096	9.849	0.2014	0.4149	0	2.188	5.008

Full table: results/tables/T08_representative_designs.csv (+ .md). Made by scripts/experiments/tables.py T08

T09. Manufacturability checks of the exported geometry (numpy mirror and PicoGK laptop builds): min wall, throat, connectivity, pockets, loose solid, volume fraction, PicoGK-vs-numpy IoU.

design	geometry	source	voxel_mm	min_wall_mm	throat_mm	connected	pockets	loose_solid	all_pass	iou_vs_numpy
end_rings_knee	30 deg sector	numpy mirror	0.07	0.3805	1.441	True	0	0	True	
end_rings_min_R	30 deg sector	numpy mirror	0.07	0.6648	0.7852	True	0	0	True	
slots_best_at_cap	30 deg sector	numpy mirror	0.06	0.3063	1.547	True	0	0	False	
task12_slots_best_at_cap_gyroid	full jacket	PicoGK (laptop)	0.2		1.219	True	0	0	True	
task12_slots_best_at_cap_gyroid	30 deg sector	PicoGK (laptop)	0.05	0.482	1.066	True	0	0	True	0.9968
task12_slots_best_at_cap_gyroid	30 deg sector	numpy mirror	0.06	0.5023	1.031	True	0	0	True	